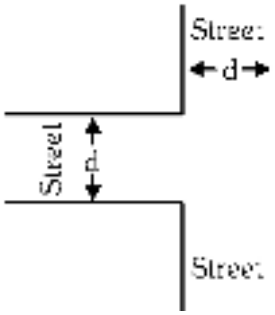
PHYSICS

1) Two long perpendicular streets of width d are connected as shown in figure. What would be the maximum length of rod which can be transported horizontally through joint of the streets ?



(1) d

(2) 2d

(3) $\sqrt{2}d$

(4) $2\sqrt{2}d$

2) $\sqrt{2+\sqrt{2+2\cos 4\theta}}$ = ... ?

(1) 2 sin θ

(2) 2 cos 2θ

(3) 2 cos θ

(4) cos θ

3) Co-ordinates of two particles is given by
$x_A = 3t$, $x_B = 15t$ and
$y_A = 2t^2$, $y_B = 7t^2$
Then slope of line AB at time t = 1sec will be :

(1) $\frac{3}{4}$

(2) $\frac{4}{3}$

(3) $\frac{5}{12}$

(4) $\frac{5}{13}$

4) $y = e^{-3x} + \log_e x^3 + \sin 2x$; $\frac{dy}{dx}$ = ... ?

(1) $e^{-3x} + \frac{1}{x^3} + \cos 2x$

(2) $3e^{-3x} + \frac{3}{x} + \cos 2x$

(3) $-3e^{-3x} + \frac{3}{x} + 2\cos 2x$

(4) None of these

5) If $y = 2x^3 - 12x^2 + 36$, then for minimum value of y is :

(1) x = 0

(2) x = 4

(3) x =1

(4) None

6) $\int \left(\frac{1}{x} + \tan^2 x\right) dx$ is equal to :-

(1) $\log_e x + \sec x.\tan x + c$

(2) $\tan x + \log_e x + c$

(3) $x + \tan x + c$

(4) $-x + \tan x + \log_e x + c$

7) The values of x for the equation $2x^2 + 5x - 12 = 0$ are :-

(1) $\frac{3}{2}, -4$

(2) $\frac{1}{2}, -2$

(3) $\frac{2}{3}, -1$

(4) $\frac{-3}{2}, 4$

8) Find the sum of given Arithmetic Progression 4 + 8 + 12 + ...+ 60 :

(1) 464

(2) 540

(3) 544

(4) 480

9) Given that $P = \frac{4}{\sqrt{4 + \frac{r^2}{2}}}$ and (r << 2), then P is roughly equal to :

(1) $\left(1 - \frac{r^2}{16}\right)$

(2) $\left(2-\frac{r^2}{8}\right)$

(3) $\left(1-\frac{r}{16}\right)$

(4) $\left(2-\frac{r}{8}\right)$

10) Let $\vec{C} = \vec{A} + \vec{B}$ then select the true statement:-

(1) $|\vec{C}|$ is always greater than $|\vec{A}|$

(2) $|\vec{C}| < |\vec{A}|$ and $|\vec{C}| < |\vec{B}|$ may be possible

(3) C is always equals to A + B

(4) C can not be equal to A + B

11) If 'a' and 'c' are scalar constants and 'b' is a unit vector, then which operation is/are possible ?

(i) acb (ii) $\frac{ab}{c}$ (iii) a + b (iv) $\frac{ac}{b}$
(v) a + b + c (vi) cb/a

(1) Only (i), (ii), (iii) and (iv) are possible

(2) Only (i), (ii) and (iv) are possible

(3) Only (i), (ii) and (vi) are possible

(4) All operations are possible

12) If vector $2\hat{i} + 2\hat{j} - 2\hat{k}$, $-\hat{i} + 2\hat{j} + 2\hat{k}$ and $5\hat{i} + y\hat{j} + \hat{k}$ are coplanar, the value of y is :-

(1) 81

(2) 36

(3) 23

(4) 9

13) If $|\hat{A} + \sqrt{2}\,\hat{B}| = |\hat{C}|$ then angle between $\hat{A}$ and $\hat{B}$ will be :

(1) 45°

(2) 60°

(3) 135°

(4) 150°

14) Find component of $\vec{A} = \hat{i} + 2\hat{j} - 3\hat{k}$ in direction of $\vec{B} = 3\hat{i} + 2\hat{j}$:-

(1) $\frac{13}{7}\left(3\hat{i} + 2\hat{j}\right)$

(2) $\frac{7}{13}\left(3\hat{i} + 2\hat{j}\right)$

(3) $\frac{7}{\sqrt{14}}\left(\hat{i}+2\hat{j}-3\hat{k}\right)$

(4) $\frac{\sqrt{14}}{7}\left(\hat{i}+2\hat{j}-3\hat{k}\right)$

15) If $\vec{p} = 3\hat{i}+4\hat{j}+5\hat{k}$ and $\vec{Q} = \hat{i}+\hat{j}$ then find a vector whose direction is same as that of $\vec{Q}$ and magnitude is same as that of $\vec{p}$

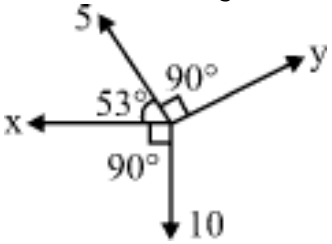
(1) $\frac{5}{\sqrt{2}}(\hat{i}+\hat{j})$

(2) $5(\hat{i}+\hat{j})$

(3) $\frac{5}{\sqrt{2}}(3\hat{i}+4\hat{j}+5\hat{k})$

(4) $\frac{3\hat{i}+4\hat{j}+5\hat{k}}{5}$

16) Find the magnitude of the unknown forces X and Y if sum of all forces is zero.



(1) 4, 8

(2) 5, 10

(3) 4, 10

(4) 5, 5

17) If a vector is making an angle α, β & γ with x, y & z axes respectively then which of the following can be possible values of α, β, & γ :-

(1) 45°, 45°, 90°

(2) 60°, 60°, 45°

(3) 53°, 90°, 37°

(4) All of these

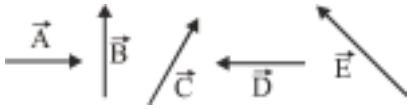
18) A particle is given successive displacements. Which of the following sets of displacements could be capable of returning the particle to its initial position ?

(1) 10 m, 8m, 6 m, 30 m

(2) 20 m, 10 m, 6m, 50 m

(3) 70m, 20 m, 40 m, 30 m

(4) 100 m, 18m, 22 m, 32 m

19) Which of the following may be possible :- 

(1) $\vec{B} + \vec{C} = \vec{A}$
(2) $\vec{B} + \vec{D} = \vec{E}$
(3) $\vec{C} + \vec{D} = \vec{A}$
(4) $\vec{A} + \vec{B} = \vec{E}$

20) ABCDEF is a regular hexagon with centre O. The value of $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} + \overrightarrow{DE} + \overrightarrow{EF} + \overrightarrow{FA}$ is

(1) $\vec{0}$
(2) $2\,\overrightarrow{AO}$
(3) $2\,\overrightarrow{AF}$
(4) $6\,\overrightarrow{AO}$

21) At what angle must the two forces (X + Y) and (X – Y) act so that the resultant may be $\sqrt{X^2 + Y^2}$?

(1) $\cos^{-1}\left(-\frac{X^2+Y^2}{2(X^2-Y^2)}\right)$
(2) $\cos^{-1}\left(-\frac{X^2-Y^2}{X^2+Y^2}\right)$
(3) $\cos^{-1}\left(-\frac{X^2+Y^2}{X^2-Y^2}\right)$
(4) $\cos^{-1}\left(-\frac{2(X^2-Y^2)}{X^2+Y^2}\right)$

22) The focal length of a concave mirror is 20 cm. Determine where an object must be placed to form an image magnified two times when the image is real :

(1) 30cm from the mirror
(2) 10cm from the mirror
(3) 20cm from the mirror
(4) 15cm from the mirror

23) **Assertion :** The focal length of a spherical mirror does not depend on wavelength of light
Reason : In case of reflection of light from denser medium the phase changes by π.

(1) Both assertion & reason are true & reason is correct explanation of assertion
(2) Both assertion & reason are true & reason is not correct explanation of assertion
(3) Assertion is true & reason is false
(4) Both assertion & reason are false.

24) An object is placed in front of a concave mirror of focal length f. A virtual image is formed with a magnification of 2. To obtain a real image of same magnification, the object has to be moved by a distance :-

(1) f

(2) f/2

(3) 3f/2

(4) 2f/3

25) A point object is placed at a distance of 10 cm and its real image is formed at a distance of 20 cm from a concave mirror. If the object is moved by 0.1 cm towards the mirror, the image will shift by about :-

(1) 0.4 cm away from the mirror

(2) 0.4 cm towards the mirror

(3) 0.8 cm away from the mirror

(4) 0.8 cm towards the mirror

26) A light ray enter in a glass slab of R.I. = $\sqrt{3}$. The angle of incident is 60°. If ray travel 5 cm inside the slab and emerge out of the slab. Find the lateral shift between the incident ray and emergent ray :-

(1) $\frac{5\sqrt{3}}{2}$ cm

(2) $\frac{7}{2}$ cm

(3) $\frac{5}{2}$ cm

(4) $\frac{5}{\sqrt{2}}$ cm

27) A vessel is quarter filled with a liquid of refractive index μ. The remaining parts of the vessel is filled with an immiscible liquid of refractive index $\frac{3}{2}$μ. The apparent depth of the vessel is 50% of the actual depth. The value of μ is :-

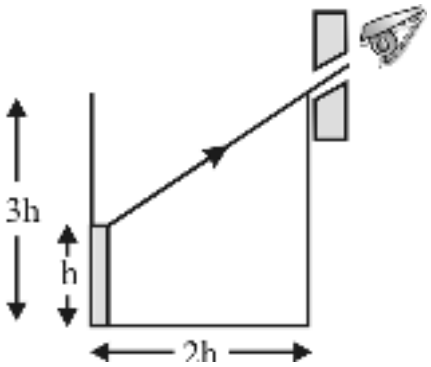
(1) 1

(2) $\frac{3}{2}$

(3) $\frac{2}{3}$

(4) $\frac{4}{3}$

28) An observer can see through a pin-hole the top end of a thin rod of height h, placed as shown in the figure. The beaker height is 3h and its radius h. When the beaker is filled with a liquid up to a height 2h, he can see the lower end of the rod. Then the refractive index of the liquid is



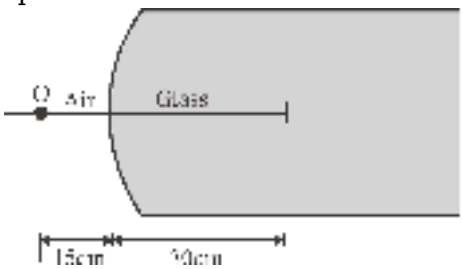
(1) 5/2

(2) $\sqrt{(5/2)}$

(3) $\sqrt{(3/2)}$

(4) 3/2

29) A point object O is placed as shown in the diagram, in front of a glass $\left(\mu = \frac{3}{2}\right)$ rod having spherical end of radius of curvature 30 cm. The image would be formed at:-



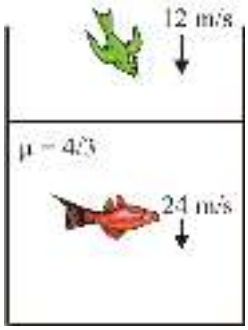
(1) 30 cm to the left of spherical surface

(2) infinity

(3) 10 cm to the right of spherical surface

(4) 18 cm to the left of spherical surface

30) A fish and a bird are moving as shown in figure. Find the velocity of bird as observed by fish.



(1) 8 m/s upwards

(2) 8 m/s downwards

(3) 9 m/s upwards

(4) 9 m/s downwards

31) An object is placed at 10 cm from the surface of a glass - slab of thickness 6 cm. Position of its

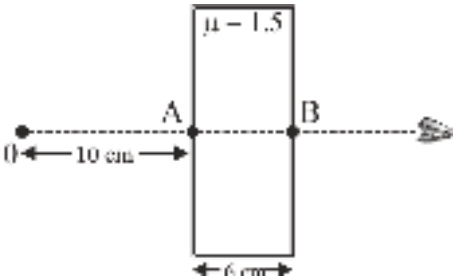


image when seen through it will be (from B) :-

(1) 2 cm

(2) 8 cm

(3) 14 cm

(4) 18 cm

32) A object is placed at 21 cm in front of concave mirror of radius of curvature of 10 cm. A glass slab of thickness 3 cm and μ = 1.5 is placed at 15 cm from mirror. If only one refraction from glass slab and reflection from mirror is considered then locate the position of image :-

(1) $\frac{23}{3}$cm

(2) 21 cm

(3) $\frac{20}{3}$cm

(4) 25 cm

33) The speed of light in media M_1 and M_2 are 1.5×10^8 m/s and 2.0×10^8 m/s respectively. A ray of light enters from medium M_1 to M_2 at an incidence angle i. If the ray suffers total internal reflection, the value of i is :-

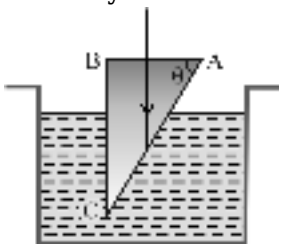
(1) equal to $\sin^{-1}\left(\frac{2}{3}\right)$

(2) equal to or less $\sin^{-1}\left(\frac{2}{5}\right)$

(3) equal to or greater than $\sin^{-1}\left(\frac{3}{4}\right)$

(4) less than $\sin^{-1}\left(\frac{2}{3}\right)$

34) A glass prism (μ = 1.5) is dipped in water (μ = 4/3) as shown in figure. A light ray is incident normally on the surface AB. It reaches the surface BC after totally reflected, if :-



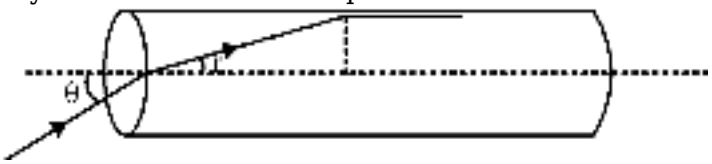
(1) $\sin\theta \geq 8/9$

(2) $2/3 < \sin\theta < 8/9$

(3) $\sin\theta \leq 2/3$

(4) It is not possible

35) A transparent solid cyclindrical rod has a refractive index of $\frac{2}{\sqrt{3}}$. It is surrounded by air. A light ray is incident at the mid-point of one end of the rod as shown in the figure.



The incident angle θ for which the light ray grazes along the wall of the rod is :-

(1) $\sin^{-1}\left(\frac{2}{\sqrt{3}}\right)$

(2) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

(3) $\sin^{-1}\left(\frac{1}{2}\right)$

(4) $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$

36) A prism (μ = 1.5) has the refracting angle of 30°.The deviation of a monochromatic ray incident normally on its one surface will be
(sin 48° 36' = 0.75)

(1) 18°36'

(2) 20°30'

(3) 18°

(4) 22°1'

37) The angle of a prism is 'A'. One of its refracting surfaces is silvered. Light rays falling at an angle of incidence 2A on the first surface returns back through the same path after suffering reflection at the silvered surface. The refractive index μ, of the prism is :-

(1) 2 sin A

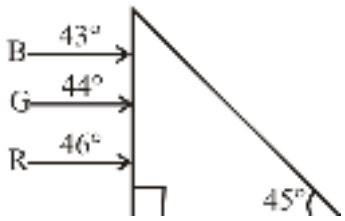
(2) 2 cos A

(3) $\frac{1}{2}\cos A$

(4) tanA

38) Figure shows a mixture of blue, green and red coloured rays incident normally on a right angled prism. The critical angles of the material of the prism for red, green and blue are 46°, 44° and 43°

respectively. The arrangement will separate.



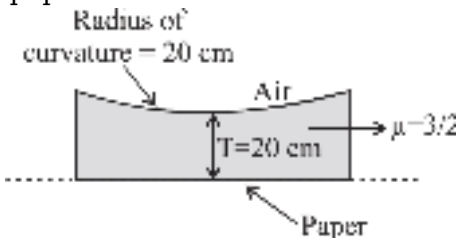
(1) Green colour from red and blue

(2) All the three colurs

(3) Red colour from blue and green

(4) Blue colour from red and green

39) A plano-concave lens is placed on a paper on which a flower is drawn. how far above its actual position does the flower appear to be ?



(1) 10 cm

(2) 15 cm

(3) 50 cm

(4) None

40) A convex lens of focal length 30 cm produces 5 times magnified real image of an object. Object distance from lens will be :-

(1) 36 cm

(2) 25 cm

(3) 30 cm

(4) 150 cm

41) The distance between an object and the screen is 80 cm. A lens produces an image on the screen when the lens is placed at either of the positions 20 cm apart. Focal length of the lens is :-

(1) 18.75 cm

(2) 20.5 cm

(3) 10.5 cm

(4) 24.5 cm

42) A concave lens of glass, refractive index 1.5, has both surfaces of same radius of curvature R. On immersion in a medium of refractive index 1.75, it will behave as a :-

(1) convergent lens of focal length 3.5R

(2) convergent lens of focal length 3.0 R.

(3) divergent lens of focal length 3.5 R

(4) divergent lens of focal length 3.0 R

43) Two thin lenses, when in contact, produce a combination of power +10 diopters. When they are 0.25 m apart, the power reduces to +6 diopters. The focal length of the lenses are

(1) 0.5 m, 0.125 m

(2) 0.1 m, 0.125 m

(3) 0.5 m, 0.5 m

(4) 0.2 m, 0.5 m

44)

Two plano-concave lenses of glass of refractive index 1.5 have radii of curvature of 20 cm and 30 cm. They are placed in contact with curved surfaces towards each other and the space between them is filled with a liquid of refractive index (4/3). Find the focal length of the system.

(1) 60 cm

(2) 72 cm

(3) 40 cm

(4) 34 cm

45) A point object is at 30 cm from a convex lens (μ = 3/2) of focal length 20 cm. The final image of object will be at infinity, if :-

(1) another concave lens of focal length 60 cm is placed in contact with the previous lens

(2) another convex lens of focal length 60 cm is placed at a distance of 30 cm from the first lens

(3) the whole system is immersed in a liquid of refractive index 4/3

(4) the whole system is immersed in a liquid of refractive index 2

CHEMISTRY

1)

N_2O_5 is decomposed according to given reaction if initial pressure of N_2O_5 is 100 torr and after 10 min total pressure of reaction mixture becomes 150 torr then rate of decomposition of N_2O_5 upto 10 min is–
$2N_2O_5 \rightarrow 4NO_2 + O_2$

(1) 3.33 torr min^{-1}

(2) 6.67 torr min^{-1}

(3) 5 torr min^{-1}

(4) 10 torr min^{-1}

2) For the reaction, $2N_2O_5(g) \rightarrow 4\ NO_2(g) + O_2(g)$, the rate of formation of O_2 is 0.032g $hour^{-1}$, then rate of conversion of N_2O_5 in g $hour^{-1}$ is :-

(1) 0.216

(2) 0.098

(3) 0.627

(4) 0.313

3) A reaction $2A + B \xrightarrow{k} C + D$ is first order with respect to A and 2nd order with respect to B. Initial conc. (t = 0) of A is C_0 while B is $2C_0$. If at t = 30 minutes the conc. of C is $C_0/4$ then rate expression at t = 30 minutes is :

(1) $R = 7\ C_0^3 k/16$

(2) $R = 27\ C_0^3 k/32$

(3) $R = 247\ C_0^3 k/64$

(4) $R = 49\ k\ C_0^3/32$

4) **Assertion (A) :-** Rate of reaction is increased in presence of catalyst.
Reason (R) :- Catalyst increases energy of reactant molecules.

(1) Both A and R are correct and R is correct explanation of A.

(2) Both A and R are correct but R is not correct explanation of A.

(3) A is correct but R is not correct

(4) A is not correct but R is correct.

5) Match the column-I with column-II

	Column-I		**Column-II**
(A)	Thermal decomposition of HI on gold surface	(P)	Zero order reaction
(B)	Inversion of cane sugar in acidic medium	(Q)	Pseudo first order reaction
(C)	Hydrogenation of ethene	(R)	First order reaction
(D)	The decomposition of gaseous ammonia on a hot platinum surface at high pressure	(S)	Second order reaction

(1) A→P, B→Q, C→S, D→P

(2) A→Q, B→P, C→S, D→R

(3) A→P, B→Q, C→R, D→P

(4) A→P, B→Q, C→R, D→S

6) A sample of gaseous aldehyde, at 518°C initially at a pressure of 363 torr, had rate of decomposition 1.00 torr s^{-1} when 5% reacted and 0.5 torr s^{-1} when 33% had reacted the order of reaction is :

(1) 3

(2) 1

(3) 0

(4) 2

7) Which of the following is a correct statement ?

(1) Molecularity & order of elementary reaction is same

(2) Activation energy is the energy difference between activated complex and energy of reactant

(3) Effective collision depends on activation energy and proper orientation.

(4) All of the above

8) Consider the reaction :
$Cl_2(aq) + H_2S(aq) \rightarrow S(s) + 2H^+(aq) + 2Cl^-(aq)$
The rate equation for this reaction is :
$Rate = k[Cl_2][H_2S]$
Which of these mechanism is/are consistent with this rate equation ?

A. $Cl_2 + H_2S \rightarrow H^+ + Cl^- + Cl^+ + HS^-$ (slow)
$Cl^+ + HS^- \rightarrow H^+ + Cl^- + S$ (fast)

B. $H_2S \rightleftharpoons H^+ + HS^-$ (fast equilibrium)
$Cl_2 + HS^- \rightarrow 2Cl^- + H^+ + S$ (slow)

(1) Neither A nor B

(2) A only

(3) B only

(4) Both A and B

9) **Assertion :** The order of a reaction can have fractional value.
Reason : The order of a reaction is given by a balanced equation

(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion

(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion

(3) The assertion is true but the reason is false

(4) The assertion is false but reason is true

10) The initial rate of reaction A + 5B + 6C → 3L + 3M has been determined under the following conditions :-

Experiment No.	**$[A]_0$ (in M)**	**$[B]_0$ (in M)**	**$[C]_0$ (in M)**	**Initial rate (in M min^{-1})**
1.	0.02	0.02	0.02	2.08×10^{-3}
2.	0.01	0.02	0.02	1.04×10^{-3}
3.	0.02	0.04	0.02	4.16×10^{-3}
4.	0.02	0.02	0.04	8.32×10^{-3}

Determine the rate law ?

(1) Rate = $k[A][B][C]^2$

(2) Rate = $k[A][C]^2$

(3) Rate = $k[A][B][C]$

(4) Rate = $k[A][B]^5[C]^6$

11) The reaction (A) $\xrightarrow{k}$ products, is zero order while the reaction (B) $\xrightarrow{k}$ products, is first order if initial concentration of A is 0.693 M then :-

(1) $(t_{1/2})_A = (t_{1/2})_B$

(2) $2(t_{1/2})_A = (t_{1/2})_B$

(3) $(t_{1/2})_A = 2(t_{1/2})_B$

(4) $4(t_{1/2})_A = (t_{1/2})_B$

12) A first order reaction undergoes 90% completion in 90 minutes, then it undergoes 99.9% completion in –

(1) 99 min

(2) 99.9 min

(3) 180 min

(4) 270 min

13) The decomposition of $NH_{3(g)}$ on platinum surface at high pressure is zero order reaction.

$$2NH_{3(g)} \xrightarrow{\Delta} N_{2(g)} + 3H_{2(g)}$$

if rate constant (k) = 2.6×10^{-3} mol L^{-1} s^{-1} then what is the rate of production of $H_{2(g)}$?

(1) 8.6×10^{-4} mol $L^{-1}s^{-1}$

(2) 7.8×10^{-3} mol $L^{-1}s^{-1}$

(3) 5.2×10^{-3} mol $L^{-1}s^{-1}$

(4) None of these

14) For any 1st order gaseous reaction A → 2B pressure developed after 20 min is 16.4 atm and after long time is 20 atm. What is the total pressure developed after 10 min.

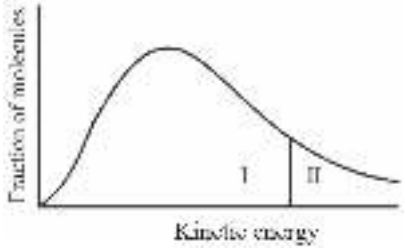
(1) 12 atm

(2) 13 atm

(3) 14 atm

(4) 15 atm

15) Consider the following graph of the kinetic energy distribution among molecules at temperature T.

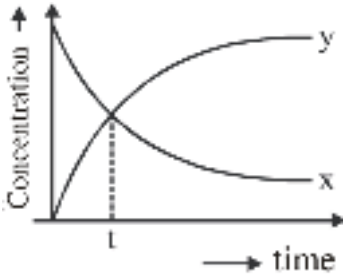


If the temperature is increased, how would the resulting graph

differ from the above one :

(1) Both area I and II would increase.
(2) Both area I and II would decrease.
(3) Area I would increase and area II would decrease
(4) Area I would decrease and area II would increase

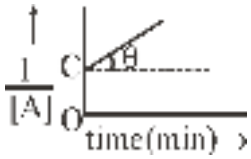
16) For the reaction x → 2y, following curve is obtained, then time 't' is :



(1) $t_{2/3}$
(2) $t_{1/4}$
(3) $t_{1/2}$
(4) $t_{1/3}$

17)

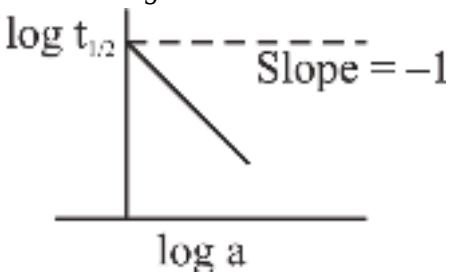
For a second order reaction, A → Product, the graph is plotted against $\frac{1}{[A]}$ and time 't'.



tan θ = 0.6 and OC = 2L mol^{-1}.The rate of reaction at begining is

(1) 1.25 mol L^{-1} min^{-1}
(2) 0.150 mol L^{-1} min^{-1}
(3) 0.5 mol L^{-1} min^{-1}
(4) 1.50 mol L^{-1} min^{-1}

18) A graph between log $t_{1/2}$ and log a, where 'a' being the initial concentration of A in the reaction



for reaction A → Product, the rate law is :-

(1) $\frac{-d[A]}{dt} = K$
(2) $\frac{-d[A]}{dt} = K[A]$

(3) $\frac{-d[A]}{dt} = K[A]^2$

(4) $\frac{-d[A]}{dt} = K[A]^3$

19)

The unit of rate of reaction & rate constant is same for :

(1) Third order reaction

(2) Second order reaction

(3) First order reaction

(4) Zero order reaction

20) The first order rate constant k is related to temp. as log k = 15.0 – (10^6 /T) Which of the following pair of value is correct ?

(1) A = 10^{15} and E = 1.9×10^4 KJ

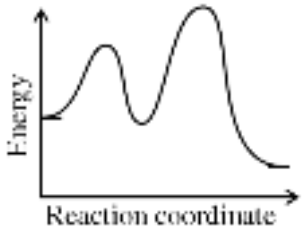
(2) A = 10^{-15} and E = 40 KJ

(3) A = 10^{15} and E = 40 KJ

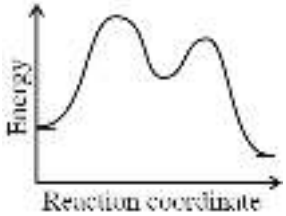
(4) A = 10^{-15} and E = 1.9×10^4 KJ.

21) Reactant R converts to product P through given mechanism (with net absorption of heat)
R → X ; slow ;
X → P ; fast ;
which of the following represent the above reaction mechanism :-

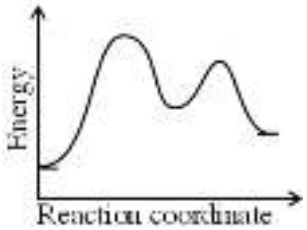
(1)

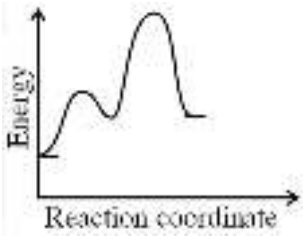


(2)



(3)



(4) 

22) If $t_{\frac{1}{2}} = \frac{2\sqrt{k_1}a^{-2}}{4k_2 + 6\sqrt{5}}$ order of reaction will be-

(1) 1

(2) 0

(3) 2

(4) 3

23) Match the Column I with Column II.

Column I		Column II	
A.	Catalyst alters the rate of reaction.	P	Cannot be fraction or zero
B.	Molecularity	Q	Proper orientation is not there.
C.	Second half-life of first order reaction.	R	By lowering the activation energy
D.	$e^{-E_a/RT}$	S	Is same as the first
E.	Energetically favourable reactions are sometimes slow.	T	Total probability is one.
F.	Area under the Maxwell - Boltzmann curve is constant.	U	Refers to the fraction of molecules with energy equal to or greater than threshold energy.

	A	B	C	D	E	F
(1)	R	P	S	U	Q	T
(2)	S	R	P	U	Q	T
(3)	P	R	S	Q	U	T
(4)	U	Q	T	S	R	P

(1) 1

(2) 2

(3) 3

(4) 4

24) Two reactions of the same order have equal Pre-exponential factors but their activation energies differ by 24.9 kJ/mol. Calculate the ratio between the rate constants $\left(\frac{K_2}{K_1}\right)$ of these reactions at 27°C :-

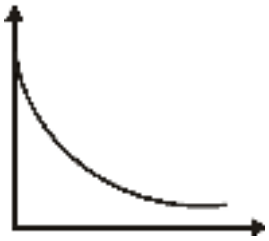
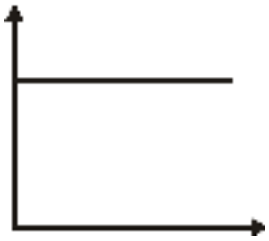
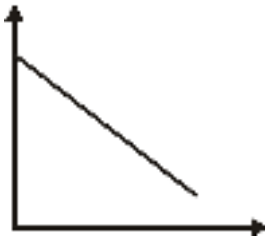
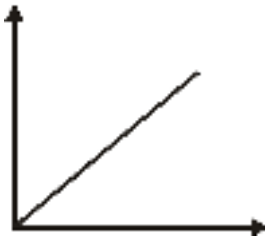
(1) 3×10^{4}

(2) $\frac{1}{3} \times 10^{-4}$

(3) $\frac{1}{3} \times 10^{4}$

(4) 3×10^{-4}

25) Match the graph in column I with the information in column II

	Column-I		**Column-II**
(i)		(A)	Rate of reaction Vs time (for zero order reaction)
(ii)		(B)	[Reactant] Vs time (for 1st order reaction)
(iii)		(C)	log [Reactant] Vs time (For 1st order reaction)
(iv)		(D)	$t_{1/2}$ Vs [Reactant] (for zero order reaction)

Here : [Reactant] is concentration of reactant

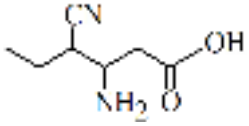
$t_{1/2}$: Half life of reaction.

(1) (i) - A, (ii) - B, (iii) - C, (iv) - D

(2) (i) - C, (ii) - A, (iii) - B, (iv) - D

(3) (i) - B, (ii) - A, (iii) - C, (iv) - D

(4) (i) - B, (ii) - A, (iii) - D, (iv) - C

26) IUPAC name of the given compound is :- 

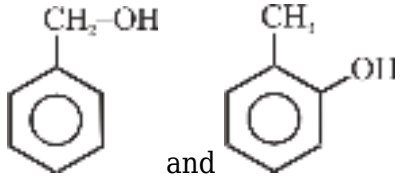
(1) 3-Amino-4-cyano hexanoic acid

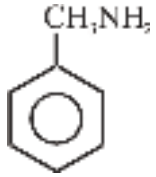
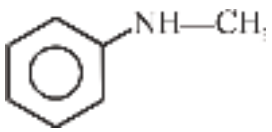
(2) 3-Cyano-4-amino hexanoic acid

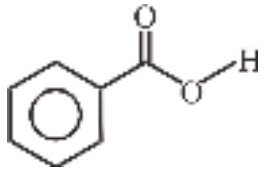
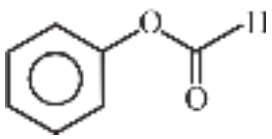
(3) 3-Amino-4-ethyl-5-cyano pentanoic acid

(4) 3-Amino-5-cyano hexanoic acid

27) Which pair of compounds does not represent same homologous series ?

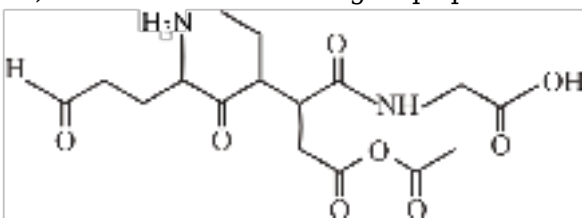
(1)  and

(2)  and 

(3)  and 

(4) All of these

28) Number of functional groups present in the following compound is :-



(1) 5

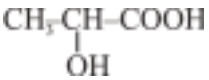
(2) 7

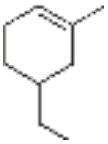
(3) 6

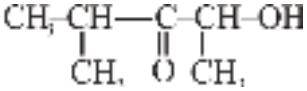
(4) 8

29) Which of the following is incorrect IUPAC name :-

(1)  3-Methyl butan-2-one

(2)  2-Hydroxy propanoic acid

(3)  5-Ethyl-1-methylcyclohex-1-ene

(4)  4-Methyl-3-oxopentan-2-ol

30) Which is incorrect match

	Common Name	IUPAC Name
(1)	Malonic acid	Propanedioic acid
(2)	Isobutyric acid	2-Methyl propanoic acid
(3)	Phenyl acetic acid	2-Phenyl ethanoic acid
(4)	Mesityl oxide	3-Methyl pent-3-en-2-one

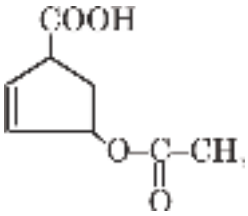
(1) 1

(2) 2

(3) 3

(4) 4

31) The correct IUPAC name of given compound is :

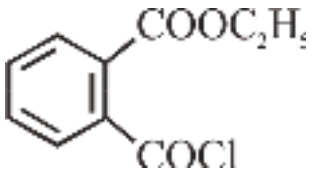


(1) 3-Ethanoyloxycyclopent-4-ene carboxylic acid

(2) 4-Ethanoyloxycyclopent-2-ene carboxylic acid

(3) 4-Ethanoyloxycyclopent-3-ene carboxylic acid

(4) 4-Ethanoyloxycyclopent-1-ene-5-carboxylic acid

32) IUPAC name of the compound  is:-

(1) 2-Chlorocarbonyl ethylbenzenecarboxylate

(2) 2-Carboxyethylbenzoyl chloride

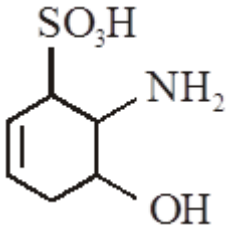
(3) Ethyl–2(chlorocarbonyl) Benzenecarboxylate

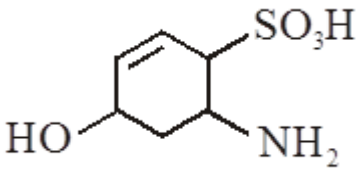
(4) Ethyl–1(chlorocarbonyl) Benzenecarboxylate

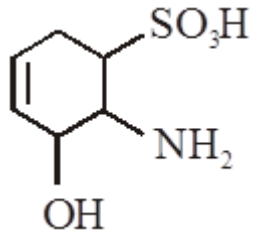
33)

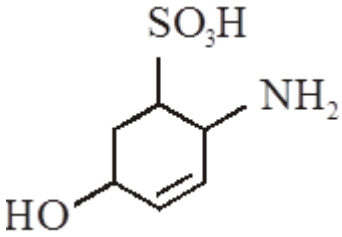
The correct structure of

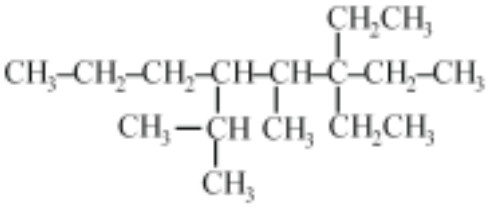
6–amino–4–hydroxycyclohex–2–ene–1–sulphonic acid is:-

(1) 

(2) 

(3) 

(4) 



34) IUPAC name of the given compound is :

(1) 3, 3-diethyl-5-isopropyloctane

(2) 3, 3-diethyl-5-isopropyl-4-methyloctane

(3) 3, 3-dimethyl-4-methyl-5-isopropyloctane

(4) None of these

35) Match the following general formula :-

(i)	C_nH_{2n-2}	(a)	Acid
(ii)	$C_nH_{2n}O_2$	(b)	Alcohol
(iii)	$C_nH_{2n+2}O$	(c)	Alkyne
(iv)	$C_nH_{2n+1}X$	(d)	Alkylhalide

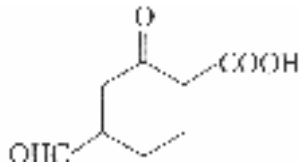
(1) a-iii, b-ii, c-i, d-iv

(2) a-ii, b-iii, c-i, d-iv

(3) a-iv, b-iii, c-i, d-ii

(4) a-ii, b-iii, c-iv, d-i

36) The correct IUPAC name of the compound :



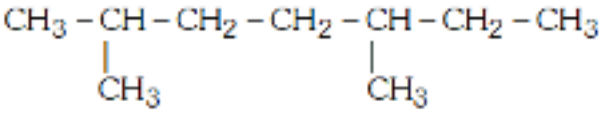
(1) 5-carboxy-3-oxocyclohexanecarboxaldehyde

(2) 2-carboxy-5-formylcyclohexane

(3) 4-formyl-2-oxocyclohexanecarboxylic acid

(4) 4-carboxy-3-oxocyclohexanal

37) **Assertion :** The name of the hydrocarbon



is 2, 5 dimethylheptane and not 3, 6-dimethylheptane.
Reason : Numbering should be done in such a way that the locants on the parent chain get lowest possible number according to lowest locant rule.

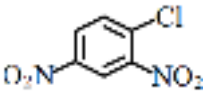
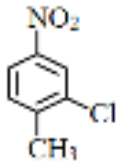
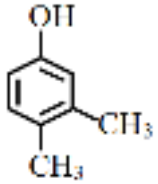
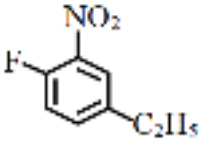
(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

38) Incorrect matched IUPAC name is :-

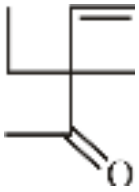
1.		1-Chloro-2,4-dinitrobenzene
2.		2-Chloro-1-methyl-4-nitrobenzene
3.		4-Hydroxy-1,2-dimethylbenzene
4.		4-Ethyl-1-fluoro-2-nitrobenzene

(1) 1

(2) 2

(3) 3

(4) 4

39) Write IUPAC name of following compound ? 

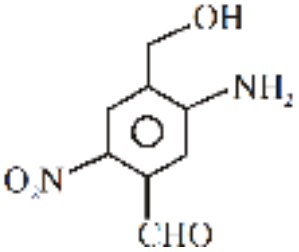
(1) 2,3-Diethylpent-1-en-2-ketone

(2) 3,3-Diethylpent-4-en-2-one

(3) 3,3-Diethylpent-1-en-1-one

(4) 3-Ethenyl-3-ethylpentane-4-one

40) The IUPAC name of the following compound is :



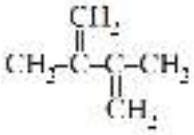
(1) 3-amino-4-hydroxymethyl-5-nitrobenzaldehyde

(2) 2-nitro-4-hydroxymethyl-5-aminobenzaldehyde

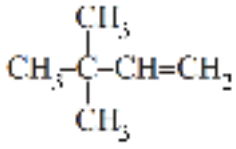
(3) 4-amino-2-formyl-5-hydroxymethylnitrobenzene

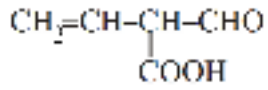
(4) 5-amino-4-hydroxymethyl-2-nitrobenzaldehyde

41) Which of the following IUPAC name is incorrect :-

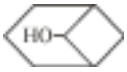
(1) 
2,3-Dimethylbuta-1,3-diene

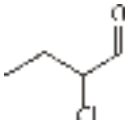
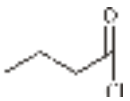
(2) $CH_3-CH=CH-C\equiv CH$
Pent-3-en-1-yne

(3) 
3,3-Dimethylbut-1-ene

(4) 
2 vinyl 3 oxopropanoic acid

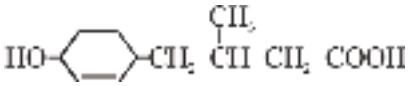
42) Consider the given statements :- (I) Acetone and Acetaldehyde are homologues

(II)  is a secondary alcohol while  is a tertiary alcohol

(III)  and  both have different functional group

Which is correct ?

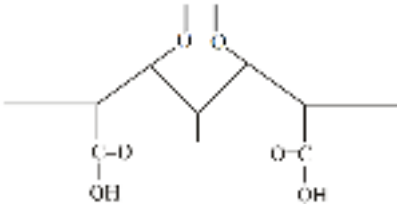
(1) I, II, III
(2) I, III
(3) II, III
(4) I, II

43) According to IUPAC system  is named as:-

(1) 4–(3–carboxy–2–methyl propyl)cyclohex–2– en–1–ol
(2) 4–(4'–hydroxy cyclohex–2'–enyl)–3-methyl butanoic acid
(3) 4–hydroxy cyclohexenyl–3–methyl butanoic acid
(4) 4–(4'–hydroxy cyclo hex–2'–ynyl)–3 methyl butanoic acid

44) Correct IUPAC name is :-

(1) 3-Methyl-2- ethylpentane
(2) 2-Ethyl- 3-methylpentane
(3) 3–Ethyl– 2-methylpentane
(4) 2-Ethyl- 2-methylpentane

45) 
What is the correct IUPAC name of the given compound :

(1) 2, 4-Dimethoxy-1,3,5-trimethylpentane-1, 5-dicarboxylic acid
(2) 2, 4, 6-Trimethyl-3, 5-dimethoxyheptane-1,7-dioic acid
(3) 3, 5-Dimethoxy-2, 4, 6-trimethylheptane-1,7-dioic acid
(4) 3, 5-Dimethoxyheptane-2,6-dicarboxylic acid

BIOLOGY

1) Which of the following is a secondary metabolite as well as a drug?

(1) Concanavalin A
(2) Vinblastine
(3) Diterpenes
(4) Ricin

2) Which of the following is **not** a homopolymer?

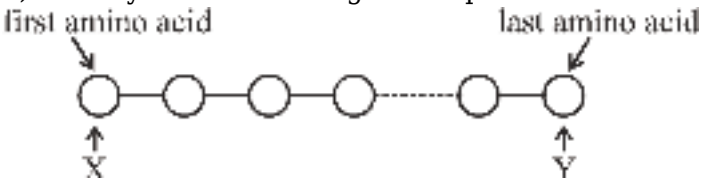
(1) Chitin

(2) Inulin

(3) Cellulose

(4) Collagen

3) Identify X and Y in the given sequence :-



	X	**Y**
1	N-terminal amino acid	C-terminal amino acid
2	N-terminal amino acid	N-terminal amino acid
3	C-terminal amino acid	N-terminal amino acid
4	C-terminal amino acid	C-terminal amino acid

(1) 1

(2) 2

(3) 3

(4) 4

4) Which of the following sets contains polysaccharides?

(1) Glucose, fructose, lactose

(2) Starch, glycogen, cellulose

(3) Sucrose, maltose, cellulose

(4) Galactose, starch, sucrose

5) **Assertion :** Iodine test can detect the presence of starch but not of cellulose in a given sample.
Reason : Starch form helical secondary structure and can hold Iodine molecules while cellulose does not have helical structure that can not hold Iodine.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

6) Which of the following is **incorrect** ?

(1) All the elements present in a sample of earth's crust are also present in a sample of living tissue

(2) The relative abundance of carbon and hydrogen with respect to other elements is higher in any living organism than in earth's crust

(3) The percentage weight of nitrogen in earth's crust is very high in comparison to human body

(4) The percentage weight of silicon in earth's crust is very high in comparison to human body

7) Match **List-I** with **List-II** :-

List-I		**List-II**	
A.	Collagen	I.	Enzyme
B.	GLUT-4	II.	Most abundant enzyme in biosphere
C.	Trypsin	III.	Most abundant protein in animal world
D.	RuBisCO	IV.	Enables glucose transport into cells.

Choose the **correct** answer from the options given below:

(1) A–II, B–III, C–I, D–IV

(2) A–III, B–IV, C–I, D–II

(3) A–I, B–III, C–IV, D–II

(4) A–III, B–I, C–IV, D–II

8) Two statements are given below :

Statement-I : Primary metabolite have identifiable functions and play known roles in normal physiological processes, we do not at the moment, understand the role or functions of all the 'secondary metabolites' in host organisms. However, many of them are useful to 'human welfare' (e.g., rubber, drugs, spices, scents and pigments)

Statement-II : The acid soluble pool represents roughly the cytoplasmic composition. The macromolecules from cytoplasm and organelles becomes the acid insoluble fraction. Together they represent the entire chemical composition of living tissues or organisms.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

9) The protein portion of an enzyme is called :

(1) Cofactor

(2) Coenzyme

(3) Apoenzyme

(4) Prosthetic group

10) The essential chemical components of many co-enzymes are:

(1) Vitamins

(2) Proteins

(3) Nucleic acids

(4) Carbohydrates

11) Enzyme catalysing rearrangement of atomic grouping without altering molecular weight or number of atoms is :

(1) Ligase

(2) Isomerase

(3) Hydrolase

(4) Oxidoreductase

12) **Assertion :** With the increase in substrate concentration the velocity of the enzymatic reaction rises at first, the reaction ultimately reaches a maximum velocity which is not exceeded by any further rise in concentration of the substrate.
Reason : Usually in any reaction enzyme molecules are more than the substrate molecules and after saturation of substrate molecules there is no free substrate molecule to bind with the additional enzyme molecules.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

13) **A.** Catalyses the transfer of any group (other than hydrogen) between a pair of substrates.
B. Catalyses removal of groups, from substrates by mechanism other than hydrolysis leaving double bond.
C. Catalyzing the linking of 2 molecules.
Identify the classes of enzymes for above functions :-

	A	**B**	**C**
(1)	Ligases	Lyases	Transferases
(2)	Transferases	Lyases	Ligases
(3)	Lyases	Ligases	Transferases
(4)	Transferases	Ligases	Lyases

(1) 1

(2) 2

(3) 3

(4) 4

14) Which one is correct?

(1) Inorganic catalysts work efficiently at high temperature and high pressure while enzymes get damaged at high temperature (above 40°C)

(2) Thermophilic organisms living in hot vents and sulphur springs have enzymes that are stable and retain their catalytic power even at high temperature (80 - 90°C)

(3) Ribozymes are nucleic acids behaving like enzyme

(4) All are correct

15) One molecule of enzyme converts the 5 molecules of substrate into product in 5 minutes. 30

molecules of substrate and two molecules of enzyme are mixed in a test tube. After 10 minutes the test tube will have :-

(1) Products and 10 catalysed molecules of substrate
(2) Only product molecules
(3) 10 uncatalysed molecules of substrate, enzymes and product molecules
(4) Products and enzymes molecules

16) Who described the first action spectrum of photosynthesis :-

(1) Julius Von Sachs
(2) T.W. Engelmann
(3) Cornelius Von Niel
(4) Jan Ingenhouz

17) The primary acceptor of CO_2 in Hatch & Slack pathway and in Calvin cycle respectively are :-

(1) RuBP & PEP
(2) PEP & PGA
(3) OAA & PGA
(4) PEP & RuBP

18) Which of the following chemicals is or are produced in light reaction of higher plants ?

(1) ATP only
(2) NADPH only
(3) ATP & NADPH
(4) ATP & NADH

19) Which of the following is formed during photorespiration?

(1) Sugar
(2) Phosphoglycolate
(3) NADPH
(4) ATP

20) Which of the following statements is **incorrect**?

(1) Cyclic photophosphorylation also occurs in grana lamellae when only light of wavelengths beyond 680 nm are available for excitation.
(2) In non-cyclic photo-phosphorylation two photosystems are connected through an electron transport chain (ETS).
(3) Lamellae of the grana have both PS I and PS II.
(4) The process through which ATP is synthesised by cells in mitochondria and chloroplasts are named as photo-phosphorylation.

21) **Assertion :-** There is not a complete one- to-one overlap between the absorption spectrum of

chlorophyll a and the action spectrum of photosynthesis.
Reason : Most of the photosynthesis takes place in the blue and red regions of the spectrum but some photosynthesis does take place at the other wavelengths of the visible spectrum.

(1) Assertion and reason both are true but reason is not the correct explanation of the assertion
(2) only Assertion is true
(3) Both assertion and reason are false
(4) Assertion and reason both are true and reason is the correct explanation of the assertion

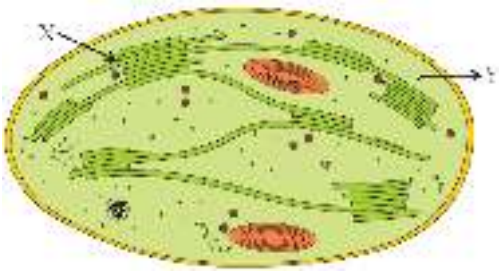
22) **Statement-I :** In respiration and photosynthesis, ATP synthesis is linked to development of a proton gradient across a membrane.
Statement-II : Plastoquinone removes a proton from the stroma to thylakoid lumen.

(1) Statement-I is correct and statement-II is incorrect.
(2) Statement-I is incorrect and statement-II is correct.
(3) Statement-I and statement-II both are incorrect.
(4) Statement-I and statement-II both are correct.

23) Match List I with List II

	List-I		**List-II**
(A)	Splitting of water molecule	(I)	Calvin pathway
(B)	Use of radioisotope of carbon	(II)	Cyclic photophosphorylation
(C)	Synthesis of only ATP	(III)	Z-scheme
(D)	Synthesis of malic acid in mesophyll cell	(IV)	Hatch and Slack pathway

Choose the correct answer from the options given below :
(1) A-I, B-II, C-III, D-IV
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-I, C-II, D-IV
(4) A-IV, B-I, C-II, D-III



24) Which one of the following correctly identify X and Y and their functions in the given figure of chloroplast?

	X		**Y**	
	Structure	**Function**	**Structure**	**Function**

(1)	Grana	Photolysis of water	Stroma	CO_2 fixation
(2)	Grana	CO_2 fixation	Stroma	Photolysis of water
(3)	Stroma	Photolysis of water	Grana	CO_2 fixation
(4)	Stroma	CO_2 fixation	Lamella	Photolysis of water

(1) 1

(2) 2

(3) 3

(4) 4

25) How many ATP are used in reduction step of calvin cycle during fixation of 6-molecule of CO_2 ?

(1) 18 ATP

(2) 6 ATP

(3) 12 ATP

(4) 30 ATP

26) The Z scheme of photophosphorylation follows the following sequence :-

PS II $\xrightarrow{A}$ e^-acceptor $\xrightarrow{B}$ ETS $\xrightarrow{B}$ PS I $\xrightarrow{C}$ e^-acceptor $\xrightarrow{D}$ $NADP^+$

Which of the following options is **correct** for A, B, C, D transfer of electrons ?

	A	**B**	**C**	**D**
(1)	Uphill	Downhill	Uphill	Downhill
(2)	Downhill	Uphill	Downhill	Uphill
(3)	Downhill	Uphill	Uphill	Downhill
(4)	Uphill	Downhill	Downhill	Uphill

(1) 1

(2) 2

(3) 3

(4) 4

27) Number of carbons in the first CO_2 acceptor and first stable product in C_4 and C_3 cycle respectively are:-

(1) 3-C, 4C , 5C, 3C

(2) 5-C, 3-C , 2-C, 4-C

(3) 4-C, 3-C , 3-C, 4-C

(4) 2-C, 3-C , 4-C, 3-C

28) **Assertion :** In light reaction of photosynthesis, splitting of water is essential during non cyclic

process.
Reason : Electrons moved from PSII to PSI, this loss can be achieved by the electrons which are released during splitting of water.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

29) Light absorption, water splitting, oxygen release, ATP formation, NADPH formation, Glucose formation, oxidation of NADPH.
How many of the above processes are carried out in photo chemical phase of photosynthesis ?

(1) 6

(2) 5

(3) 4

(4) 7

30) Given below are two statements :
Statement-I : The O_2 evolved by green plant comes from H_2O, was proved using radio isotope techniques.
Statement-II : Cornelius van Niel demonstrated that photosynthesis is essentially a light-dependent reaction in which hydrogen from a suitable oxidisable compound reduce carbon-dioxide to carbohydrate.
Choose the correct option :

(1) Statement I and statement II both are correct

(2) Statement I and statement II both are incorrect

(3) Only statement I is correct

(4) Only statement II is correct

31) Which of the following is 5C compound of Kreb's cycle:

(1) α-ketoglutaric acid

(2) Isocitric acid

(3) Cis aconitic acid

(4) None of these

32) In the electron transport system, the final acceptor of hydrogen is

(1) Cytochrome b

(2) Cytochrome a_3

(3) Oxygen

(4) Ubiquinone

33) During glycolysis :-

(1) ATP molecules are synthesized between fructose-6-phosphate and fructose 1,6-bisphosphate

(2) Water molecules are released between PEP and pyruvate

(3) NAD^+ is oxidized between PGAL and 1-3 Bi PGA

(4) ATP molecules are synthesized in the conversion of BPGA to PGA

34) The value of R.Q. during tripalmitin oxidation :

(1) 1

(2) 0.7

(3) 1.3

(4) 0

35) During anaerobic respiration H-donor is

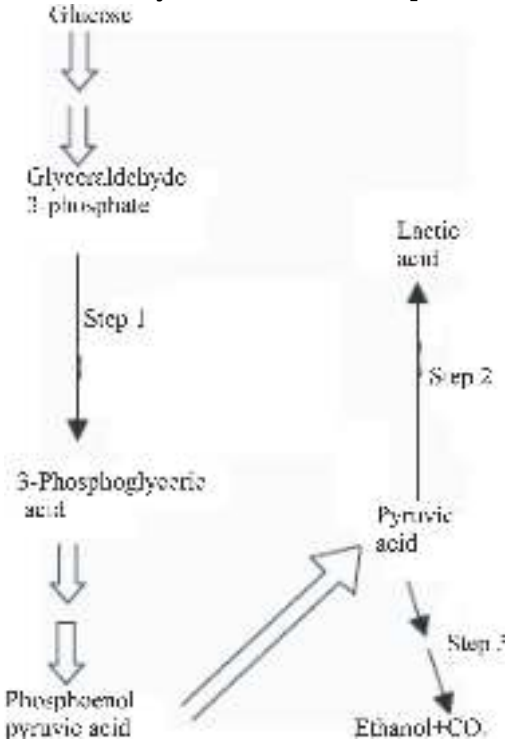
(1) PGA

(2) PGAL

(3) PEP

(4) Pyruvic Acid

36) Pathways of anaerobic respiration are shown in the diagram. Select the correct option :-



	Synthesis of NADH + H	**Utilisation of NADH + H**
1	Step-2	Step-1 & 3
2	Step-1	Step-2 & 3
3	Step-2 & 3	Step-1
4	Step-1 & 3	Step-2

(1) 1

(2) 2

(3) 3

(4) 4

37) **Assertion :** 4 ATP molecules are directly synthesized in glycolytic pathway from one glucose molecule.
Reason : 1 glucose molecule is broken down into four molecules of glyceraldehyde-3-phosphate (G3P). From each molecule of G3P converting into pyruvate produces 2 ATP.

(1) Assertion is correct but reason is incorrect.

(2) Assertion is incorrect but reason is correct.

(3) Both Assertion & Reason are true and the Reason is a correct explanation of the Assertion.

(4) Both Assertion & Reason are true and the Reason is not a correct explanation of the Assertion.

38) Identify the incorrect statement :

(1) The TCA cycle starts with the condensation of acetyl group with OAA and water to yield citric acid

(2) During the conversion of succinyl CoA to succinic acid a molecule of GTP is synthesised

(3) In ETS, O_2 acts as the final hydrogen acceptor

(4) When protein are respiratory substrate the R.Q. would be about 1.2

39) Which among the given enzymes are responsible for conversion of pyruvic acid to ethanol and CO_2 :-
Hexokinase, Pyruvic acid decarboxylase, Pyruvate kinase, Alcohol dehydrogenase, Lactate dehydrogenase.

(1) 2

(2) 3

(3) 4

(4) 5

40) Match the column I and column II:

	Column I		**Column II**
a.	2C compound	i.	Citric acid
b.	3C compound	ii.	Pyruvic acid
c.	4C compound	iii.	Malic acid
d.	5C compound	iv.	Acetyl CoA
		v.	α-Ketoglutaric acid

(1) a-iv, b-iii, c-ii, d-v

(2) a-iv, b-ii, c-iii, d-i

(3) a-iv, b-ii, c-iii, d-v

(4) a-ii, b-iii, c-iv, d-i

41) **Statement-I :-** The fructose 1,6-bisphosphate splits into dihydroxyacetone phosphate and 3-phospho-glyceraldehyde in glycolysis
Statement-II :- NADH produced in the mitochondrial matrix during citric acid cycle are oxidised by NADH dehydrogenase enzyme of E.T.S.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct.

42) Which one is correct sequence occuring in glycolysis

(1) G-6-P → PEP → 3-PGAL → 3-PGA

(2) G-6-P → 3-PGAL → 3-PGA → PEP

(3) G-6-P → PEP → 3-PGA → 3-PGAL

(4) G-6-P → 3-PGA → 3-PGAL → PEP

43) If 10 molecule of water are produced in ETS of mitochondria than find number of ATP produced and $NADH_2$ are oxidised :-

(1) 30 ATP, 10 $NADH_2$

(2) 10 ATP, 10 $NADH_2$

(3) 15 ATP, 5 $NADH_2$

(4) 6 ATP, 6 $NADH_2$

44)

Name of complexes	Component of complexes
NADH dehydrogenase	FMN, Fe-S
.....i.....	FAD, Fe-S
.....ii.....	Cytochrome b-cyt c_1 Fe-S
Cytochrome C oxidase	.....iii.....
ATP synthase	f_0-f_1

(1) i-cytochrome bc_1 complex, ii-Succinate dehydrogenase, iii-cyt a and cyt a_3

(2) i-cytochrome bc_1 oxidase, ii-fumarate dehydrogenase, iii-cyt c and cyt a_3

(3) i-Succinate dehydrogenase, ii-cytochrome bc_1 complex, iii-cyt a and cyt a_3, 2 Cu centres

(4) i-Fumarate dehydrogenase, ii-cytochrome bc_1 reductase, iii-cyt a and cyt a_3

45)

Aerobic respiration is the highly efficient metabolic process of breaking down glucose in the presence of oxygen to produce energy (ATP) CO_2 and water. It occurs in the cytoplasm and mitochondria of eukaryotic cells.
Given below are two statements about this process.

Statement-I :- The fate of pyruvic acid depends on the availability of O_2 and cellular need.
Statement-II :- There is gain of 38 ATP for two molecules of glucose.

(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

46) Closure of stomata is stimulated by

(1) Gibberellin
(2) Abscisic acid
(3) Ethylene
(4) Auxin

47) Which of the following is the characteristic feature of the cells in the phase of maturation?

(1) New cell wall deposition
(2) Cell enlargement
(3) Protoplasmic modifications
(4) Increased vacuolation

48) Natural cytokinin was isolated from :-

(1) Human urine
(2) Rice seedling
(3) Ripening fruits
(4) Corn-kernel

49) If a gardner wants to make plants short & dense then besides decapitation process which of the following plant growth regulators may helpful for this purpose ?

(1) Gibberellins
(2) Auxins
(3) Cytokinins
(4) Ethylene

50) Length of grapes stalks increase due to :-

(1) Auxin
(2) Gibberellins
(3) Cytokinins
(4) Ethylene

51) **Assertion:** The final structure at maturity of a cell or tissue is determined by the location of the cell within the plant body.
Reason: Within the plant body the cell or tissues arising out of the same meristem have different

structure at maturity.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

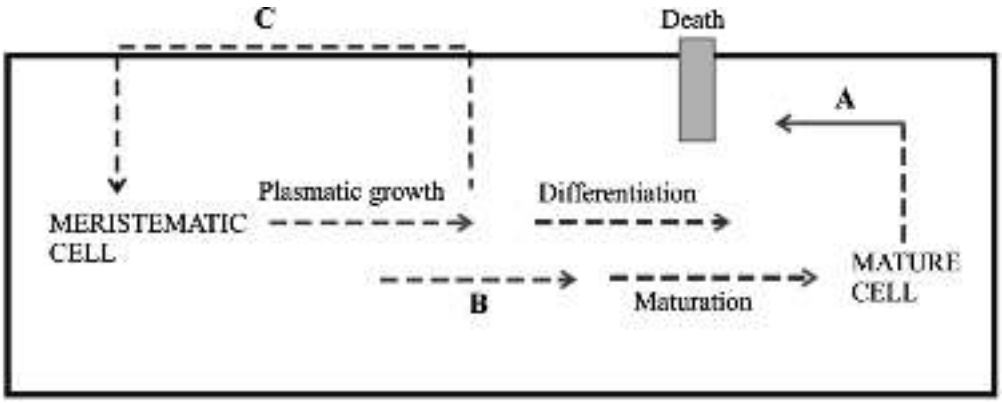
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

52)

The figure below shows three phases (A,B,C) of developmental process in a plant cell.
Select the option giving correct identification together with what it represents :-



(1) C-In this phase cells have only secondary cell wall to maintain their shape constant

(2) A-This phase of development is controlled only by intracellular intrinsic factor.

(3) B-In this phase decreased vacuolation is required to make some space for other cell organelles in cell

(4) A-This phase of development is also promoted by the PGR which is effective in fruit ripening

53) Choose the incorrect statement-

(1) Growth, differentiation and development are very closely related events in the life of a plant

(2) Development in plants under the control of intrinsic and extrinsic factors

(3) Heterophylly in cotton, coriander and larkspur are examples of plasticity

(4) Formation of interfascicular cambium and cork cambium from fully differentiated parenchyma cells is termed as redifferentation

54) Match list-I with list-II

list-I		**list-II**	
A.	Ethylene	I.	Apical dominance
B.	Cytokinins	II.	Promotion of senescenence
C.	ABA	III.	Delaying leaf Senescenence
D.	Auxins	IV.	Inhibition of seed germination

Choose the correct answer from the options given below :

(1) A-II,B-III,C-IV,D-I

(2) A-IV,B-II,C-I,D-III

(3) A-III,B-IV,C-I,D-II

(4) A-III,B-I,C-IV,D-II

55) **Statement-I :-** 2, 4-D is widely used to kill dicot weeds, does not effect mature monocot plants.
Statement-II :- Seed less fruits can be produced by spray of IAA. Eg.- Tomato.

(1) Statement I is true but statement II is false

(2) Statement I is false but statement II is true

(3) Both statement I and statement II are true

(4) Both statement I and statement II are false

56) Ethylene, GA_3, IAA, IBA, NAA, cytokinin, kinetin, ABA, Auxin
How many of the above are growth promotor hormones.

(1) 9

(2) 8

(3) 6

(4) 7

57) Which of the following statement is incorrect ?

(1) Ethylene promotes senescence and abscission of plant organs especially of leaves and flowers

(2) Ethylene breaks seed and bud dormancy, initiates germination in peanut seeds

(3) Ethylene promotes male flowers in cucumbers there by in increasing the yield

(4) Ethylene is highly effective in fruit ripening

58) The PGRs of a certain group play an important role in plant responses to stresses. How many PGRs in the list given below belongs to this group?
Indole Butyric acid, Inhibitor B, Gibberellic acid, Abscission II, Kinetin, Dormin, Naphthalene acetic acid

(1) Six

(2) Three

(3) Five

(4) One

59) **Assertion (A) :** For any and very phase of growth, and differentiation in plants plant growth regulator have some role to play such roles could be complimentary or antagonistic.
Reason (R) : Some plant growth regulators can help each other in any part of plant development and also they can overcome the effect of another plant growth regulates like auxin and cytokinin respectively in root shoot differentiation and in apical dominance.

(1) **Both Assertion** and **Reason** are correct and **Reason** is correct explanation of **Assertion**.

(2) Both **Assertion** and **Reason** are correct but **Reason** is not correct explanation of **Assertion**.

(3) **Assertion** is correct **Reason** is incorrect

(4) **Assertion** is incorrect **Reason** is correct

60) Interestingly, the discovery of each of the five major groups of PGRs have been accidental. All this started with the observation of charles Darwin and his son Francis Darwin when they observed that the coleoptiles of canary grass responded to unilateral illumination by growing towards the light source (phototropism). After a series of experiments, it was concluded that the tip of the coleoptile was the site of transmittable influence that caused the bending of entire coleoptile. The growth of coleoptile towards the source of light is because of :

(1) High conc. of auxin towards illuminated side

(2) More growth of coleoptile towards illuminated side

(3) Low concentration of auxin towards shaded side

(4) High growth of coleoptile towards shaded side.

61) pH inside lysosome is:

(1) Decreased by Active pumping of H^+ from cytoplasm

(2) Increased by Active pumping of H^+ from cytoplasm

(3) Decreased by simple diffusion of H^+ from cytoplasm

(4) Increased by simple diffusion of H^+ from cytoplasm

62) Higher plant cells and animal cells are different in how many aspect :-
(A) Plastids (B) Large central vacuole
(C) Centriole (D) Stored food
(E) Cell wall (F) Ribosome
(G) Mitochondria

(1) Five

(2) Six

(3) Three

(4) Two

63) Microtubules and microfilaments are proteinaceous filamentous structures form cytoskeleton of the cell. Which one is not a function of cytoskeleton ?

(1) Mechanical support

(2) Motility

(3) Protein synthesis

(4) Maintenance of the shape of the cell

64) Which is incorrect for golgi apparatus ?

(1) It has distinct convex (cis) and concave (trans) face

(2) Its part of endomembrane system

(3) Its cis and trans faces are entirely different and are not connected

(4) Principally it perform function of packaging of materials

65) Following are some events of mitotic cell division, select the correct order in which these events occur.

i. Condensation of chromosomes is completed.
ii. Chromosomes lose their identity as discrete elements.
iii. Sister chromatids separate.
iv. Initiation of the assembly of the mitotic spindle.

(1) i → ii → iii → iv
(2) iv → i → iii → ii
(3) ii → i → iv → iii
(4) iv → iii → i → ii

66) Which statement is incorrect about cell membrane:

(1) One of the most important functions of the plasma membrane is the transport of the molecules across it
(2) Nature of membrane lipid is quasi fluid
(3) The detail structure of the membrane was studies only after the Advent of the simple microscope
(4) The cell membrane is mainly composed of lipid and proteins

67) Choose the incorrect pair :-

(1) Golgi bodies - Densely stained reticular structure near the nucleus
(2) Cisternae - Circular, fixed in number
(3) Forming face - Convex *cis*
(4) Maturing face - Concave *trans*

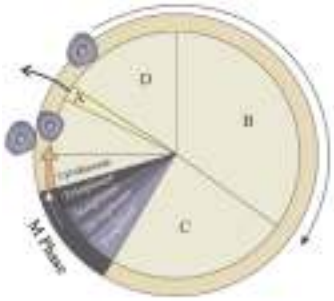
68) The stroma of the chloroplast contains enzymes required for the synthesis of

(1) Both carbohydrates and proteins
(2) Only carbohydrate
(3) Only Proteins
(4) Both Carbohydrates and Lipid

69) Lipids are arranged within the membrane with

(1) polar heads toward inner side and the hydrophobic tails toward outerside
(2) both heads and tails toward outerside
(3) heads toward outerside and tail towards inside
(4) both heads and tails toward innerside.

70) Given below is diagrammatic view of cell cycle indicating formation of two cell from one cell.



Mark the correct one for the events of this cell cycle :-

(1)	A	G_0	Cells is metabolically inactive
(2)	B	S	Amount of DNA per cell doubles but chromosome number remain same
(3)	C	G_2	Interval between mitosis and initiation of DNA replication where cell continuously grows
(4)	D	G_1	Shortest phase of cell cycle in which most of the organelles duplicate

(1) 1

(2) 2

(3) 3

(4) 4

71) Read the following four statements (A - D) :-
(A) The interphase lasts more than 95% of the duration of cell cycle.
(B) In animal cells during the s-phase, DNA replication begins in the nucleus and the centriole duplicates in the nucleolus
(C) Since the number of chromosomes in the parent and progeny cells is the same so mitosis is also called as equational division.
(D) Mitosis usually results in the production of diploid daughter cells with identical genetic complement.
How many of the above statement/s is/are incorrect?

(1) One

(2) Two

(3) Four

(4) Three

72)

Read the following four statements (a-d)
(a) Late prophase → Golgi bodies, ER disappear
(b) Metaphase → Chromosomes arrange on equator
(c) Anaphase → Centromere split

(d) Telophase → Centriole divide
How many are correctly matched ?

(1) Three

(2) Two

(3) One

(4) Four

73) In a diploid cell before s-phase quantity of DNA is 20 pico gram after anaphase-II situation will be in each daughter nucleus

(1) Diploid 10 pg DNA

(2) Haploid 10 pg DNA

(3) Haploid 5 pg DNA

(4) Diploid 5 pg DNA

74)

How many of the following statements are true?
(i) Chromatin contains DNA, RNA, histones and non-histones
(ii) A single human chromosome has approximately two meter long thread of DNA
(iii) Every chromosome essentially has a primary constriction
(iv) Nucleolus is a site for active ribosomal RNA synthesis

(v) The outer membrane of nucleus do not bear ribosomes on it

(1) Two

(2) Three

(3) Four

(4) Five

75) Read the following statement carefully and identify the correct statement and select the correct option given below :-
(A) DNA Replication occurs in nucleus in eukaryotes
(B) Gap two phases is also called post mitotic phase
(C) Crossing over in pachytene occurs due to chiasma formation
(D) All cells are arises from pre existing cell by cell division

(1) Only A and D

(2) Only A, B and D

(3) Only A, C and D

(4) A, B, C, and D

76) The morphological features of stem which are used to differentiate stem from root ?

(1) Presence of nodes and internodes

(2) Multicellular hair

(3) Positively phototropic

(4) All of the above

77) Leaf is ..... a ... outgrowth of stem developed ....b.... at the ....c..

(1) a : Terminal b: Exogenously c : Node

(2) a : Lateral : b: Exogenously c: Internode

(3) a : Lateral b : Exogenously c : Node

(4) a : Lateral b : Endogenously c : Node

78) **Statement A :** Flowers exhibit enormous variation in structure, symmetry, position of ovary in relation to other parts, arrangement of petals, sepals, ovules etc.
Statement B : Flowers can be actinomorphic, zygomorphic or asymmetric on the basis of position of ovary with regards to other part of flowers e.g. speals, petals and stamens.

(1) Statement A and B both are true
(2) Statement A and B both are false
(3) Only statement A is true
(4) Only statement B is true

79) Which of the following can be described by the floral formula and floral diagram both ?

(1) Position of ovary
(2) Floral symmetry
(3) Cohesion and adhesion within parts of whorls and between whorls
(4) Aestivation and placentation

80) How many of the following have flowers with superior ovary ?
Cucumber, sunflowers, tomato, potato, petunia, mustard, chinarose, rose, plum, mango, apple, coconut, peach

(1) Six
(2) Seven
(3) Nine
(4) Five

81) Which of the unique characteristic of angiosperms does not present in other group of plants?

(1) Fruit and Seed
(2) Flower and Fruit
(3) Flower, Fruit and Seed
(4) Flower and Seed

82) **Assertion (A) :** Stamens are epiphyllous in flowers of lily plant and epipetalous in flowers of tomato plants.
Reason (R) : Stamens in flowers of lily and tomato plants are attached to petals and tepals respectively.

(1) A and R both are true and R explain A
(2) A and R both are false
(3) A is true but R is false
(4) A and R both are true and R does not explain A

83) Which of the following is not a characteristic of potato family

(1) Valvate aestivation
(2) Actinomorphic flowers

(3) Pentamerous flower

(4) Non endospermic seed

84) How many of the following can be used to differentiated dicot plant from monocot plants ?
(a) Pentamerous flowers
(b) Ring arrangement of vascular bundles
(c) Conjoint vascular bundle in stem
(d) Radial vascular bundles in root
(e) Nature of hypodermis in stem

(1) Three

(2) Four

(3) Two

(4) One

85) Cells of meristematic region of root tip :

(1) Are differentated cell

(2) Have thick secondary cell wall

(3) Have dense cytoplasm

(4) Are large in size

86) The parenchymatous cells which lie in between xylem and phloem in dicot root is known as :

(1) Primary medullary rays

(2) Interfascicular cambium

(3) Intrafascicular cambium

(4) Conjuctive tissue

87) Two to four xylem bundle are commonly observed in :

(1) Monocot root

(2) Dicot stem

(3) Dicot root

(4) Monocot stem

88) **Statement A :** The parallel venation in monocot leaves is reflected in the different size of vascular bundles in leaves
Statement B : Size of vascular bundle in leaf depends on leaf thickness

(1) Statement A and B both are true

(2) Statement A and B both are false

(3) Only statement A is true

(4) Only statement B is true

89) **Assertion (A) :** Outermost layer of cortex in dicot stem is made up of collenchyma and is also known as starch sheath.

Reason (R) : Outermost layer of cortex is known as hypodermis and it provide mechanical support to young aerial herbaceous dicot stem.

(1) A and R both are true and R explain A

(2) A and R both are false

(3) A is false but R is true

(4) A and R both are true and R does not explain A

90) What is the position of palisade parenchyma and phloem in dicot leaves?

(1) Abaxial and adaxial respectively

(2) Adaxial and adaxial respectively

(3) Abaxial and abaxial respectively

(4) Adaxial and abaxial respectively

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	3	3	3	2	4	1	4	2	2	3	3	3	2	2	2	4	3	2	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	1	2	1	1	3	2	2	1	1	3	3	3	1	2	1	2	3	1	1
Q.	41	42	43	44	45															
A.	1	1	1	2	1															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	1	4	3	3	4	4	2	3	1	2	4	2	3	4	4	2	3	4	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	4	1	1	3	1	4	3	4	4	2	3	2	2	2	3	1	3	2	4
Q.	86	87	88	89	90															
A.	4	3	2	3	3															

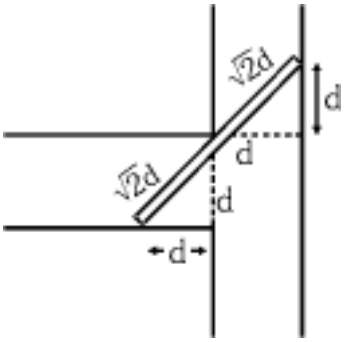
BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	4	1	2	1	3	2	1	3	1	2	3	2	4	3	2	4	3	2	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	4	3	1	3	1	1	1	2	1	1	3	4	2	2	2	1	4	1	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	2	1	3	3	2	3	4	3	2	2	4	4	1	3	4	3	2	1	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	1	1	3	3	2	3	2	1	3	2	1	1	2	2	1	4	3	3	3	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	2	3	4	1	3	4	3	3	3	4										

SOLUTIONS

PHYSICS

1)



So maximum length of rod = $2\sqrt{2}\,d$

2) $\sqrt{2+\sqrt{2(1+\cos 4\theta)}}$

$= \sqrt{2+\sqrt{2\,(1+\cos^2 2\theta)}}$

$= \sqrt{2+2\cos 2\theta}$

$= \sqrt{2(1+\cos 2\theta)}$

$= \sqrt{2(2\cos^2\theta)}$

$= 2\cos\theta$

3)

$$\text{Slope of line AB} = \frac{y_B - y_A}{x_B - x_A}$$

$$= \frac{7t^2 - 2t^2}{15t - 3t}$$

$$= \frac{5t^2}{12t} = \frac{5t}{12}$$

at t = 1 sec

$$\text{slope} = \frac{5}{12}$$

4) $y = e^{-3x} + \log_e x^3 + \sin 2x$

$$\frac{dy}{dx} = -3e^{-3x} + \frac{3}{x} + 2\cos 2x$$

5)

$y = 2x^3 - 12x^2 + 36$

$$\frac{dy}{dx} = 6x^2 - 24x = 0$$

$6x^2 - 24x = 0$
$6x^2 = 24x$
$6x\,(x - 4) = 0$
$x = 0, x = 4$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}(6x^2 - 24x)$$

$$\frac{d^2y}{dx^2} = 12x - 24$$

at $x = 4$, $\frac{d^2y}{dx^2} = 24 > 0$

$x = 4$ is point for local minima

6) $$y = \int \left(\frac{1}{x} + \tan^2 x\right) dx$$

$$y = \int \frac{1}{x} dx + \int \left(\sec^2 x - 1\right) dx$$

$y = \log_e x + \tan x - x + c$

7) $2x^2 + 5x - 12 = 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-5 \pm \sqrt{25 - 4(2)(-12)}}{2 \times 2}$$

$$= \frac{-5 \pm 11}{4} = -4 \text{ and } \frac{3}{2}$$

8) $$n = \frac{\ell}{d}, \quad S_n = \frac{n}{2}(a + \ell)$$

$a = 4, d = 4, \square = 60$

$$n = \frac{60}{4} = 15$$

$$S_{15} = \frac{15}{2}(4 + 60) = 15 \times 32 = 480$$

9) 1. Rewrite the expression: $$P = 4\left(4 + \frac{r^2}{2}\right)^{\frac{-1}{2}}$$

2. Factor out 4 from the square root:

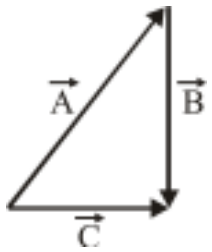
$$P = \frac{4}{\sqrt{4}\left(1 + \frac{r^2}{8}\right)^{\frac{1}{2}}} = 2\left(1 + \frac{r^2}{8}\right)^{\frac{-1}{2}}$$

3. Since $r << 2$, use $(1 + x)^n \approx 1 + nx$:

$$P \approx 2\left[1 + \left(-\frac{1}{2}\right)\left(\frac{r^2}{8}\right)\right]$$

$$P \approx 2\left(1 - \frac{r^2}{16}\right) = 2 - \frac{r^2}{8}$$

4. Simplify:

10) 

 $\vec{C}$ may be less than $\vec{A}$ and $\vec{B}$

11)

Explain Question : Which operation among the given option are correct, if a, c → scalar constant and b → unit vector

Concept : (1) Multiplication or division by a scaler and unit vector.
(2) Addition of vectors.

Solution : We can't add scalars and vectors and also can't divide by vector.
(i) a c b → It is correct (scalar constant Multiplied by vector)
(ii) $\frac{ab}{c}$ → correct
(iii) a + b → Not correct (unit vector cannot be added to scalar constant)
(iv) $\frac{ac}{b}$ → Not correct,
(v) a + b + c → Not Correct
(vi) $\frac{cb}{a}$ → Correct

Final Answer : option (3) Only (i), (ii) and (vi) are possible

12)

QUESTION EXPLANATION: Vectors $\vec{A} = 2\hat{i} + 2\hat{j} - 2\hat{k}$, $\vec{B} = -\hat{i} + 2\hat{j} + 2\hat{k}$, and $\vec{C} = 5\hat{i} + y\hat{j} + \hat{k}$ are coplanar. We need to find the value of y.

CONCEPT : Condition of Coplanarity (Scalar Triple Product = 0)

FORMULA : $\left(\vec{A} \times \vec{B}\right).\vec{C} = 0$

SOLUTION : $\vec{A} \times \vec{B} = \left(2\hat{i} + 2\hat{j} - 2\hat{k}\right) \times \left(-\hat{i} + 2\hat{j} + 2\hat{k}\right) = \vec{D}\,(\text{say})$
$\vec{D} = 8\hat{i} - 2\hat{j} + 6\hat{k}$
$\vec{C} = 5\hat{i} + y\hat{j} + \hat{k}$
Since given three vector are coplanar $\vec{C}$ and $\vec{D}$ are ⊥ to each other.
Therefore, $\vec{C}.\vec{D} = 0$
40 – 2y + 6 = 0
y = 23

Final Answer : Option (3)

13) **Explain Question:** In this question, we are asked that if $|\hat{A} + \sqrt{2}\,\hat{B}| = |\hat{C}|$ then what is the angle between $\hat{A}$ and $\hat{B}$, assuring all are unit vectors.

Concept: Addition of vectors.

Solution:

Magnitude of sum of two vectors is given by $\left|\vec{A} + \vec{B}\right| = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

$|\vec{A}| = |\hat{A}| = 1, \quad |\vec{B}| = 2|\hat{B}| = \sqrt{2} \times 1 = \sqrt{2}$

$|\vec{A} + \vec{B}| = \left|\hat{A} + \sqrt{2}\hat{B}\right| = \left|\hat{C}\right| = 1$

$$\left|\hat{A} + \sqrt{2}\hat{B}\right| = \sqrt{\left|\hat{A}\right|^2 + \left|\sqrt{2}\hat{B}\right|^2 + 2\left|\hat{A}\right|\left|\sqrt{2}\hat{B}\right|\cos\theta}$$

$$\Rightarrow (1)^2 = (1)^2 + \left(\sqrt{2}\right)^2 + 2 \times 1 \times \sqrt{2} \times \cos\theta$$

$$\Rightarrow 1 = 1 + 2 + 2\sqrt{2}\cos\theta$$

$$\cos\theta = -\frac{1}{\sqrt{2}}$$

$$\therefore \theta = 135^\circ$$

Final Answer: Option (3)

14) **Explain Question:** Find the vector component (projection) of $\vec{A}$ in the direction of $\vec{B}$.

Concept: Projection of a vector on another vector (Application of dot product)

Solution: Vector projection formula:

Component of $\vec{A}$ along $\vec{B}$ $= \dfrac{\vec{A} \cdot \vec{B}}{\left|\vec{B}\right|^2}\vec{B}$

Given

$\vec{A} = \hat{i} + 2\hat{j} - 3\hat{k}, \vec{B} = 3\hat{i} + 2\hat{j}$

Dot product : $\vec{A} \cdot \vec{B} = (1)(3) + (2)(2) + (-3)(0) = 3 + 4 = 7$

Magnitude squared of $\vec{B}$

$\left|\vec{B}\right|^2 = 3^2 + 2^2 = 9 + 4 = 13$

Projection vector : Component of $\vec{A}$ along $\vec{B} = \dfrac{7}{13}\left(3\hat{i} + 2\hat{j}\right)$

Final answer: Option (2)

15) **Explain the Question :** In this question, we need a vector that has direction same as $\vec{Q} = \hat{i} + \hat{j}$.

Magnitude same as $\vec{P} = 3\hat{i} + 4\hat{j} + 5\hat{k}$.

Concept : This question is based on magnitude of vector and unit vector.

Solution : A vector with magnitude $|\vec{P}|$ and direction of $\vec{Q}$ is

$$\vec{R} = |\vec{P}| \times \frac{\vec{Q}}{|\vec{Q}|}$$

Step-1

Magnitude of $\vec{P}$

$$|\vec{P}| = \sqrt{(3)^2 + (4)^2 + (5)^2} = \sqrt{50} = 5\sqrt{2}$$

Step-2

Magnitude of $\vec{Q}$

$$|\vec{Q}| = \sqrt{(1)^2 + (1)^2} = \sqrt{2}$$

Step-3 : Unit vector along $\vec{Q}$

$$\hat{Q} = \frac{\vec{Q}}{|\vec{Q}|} = \frac{\hat{i} + \hat{j}}{\sqrt{2}}$$

$$\vec{R} = |\vec{P}| \times \frac{\vec{Q}}{|\vec{Q}|}$$

Step-4 : Required vector,

$$\vec{R} = 5\sqrt{2} \times \frac{(\hat{i} + \hat{j})}{\sqrt{2}} = 5(\hat{i} + \hat{j})$$

Final Answer : Option (2)

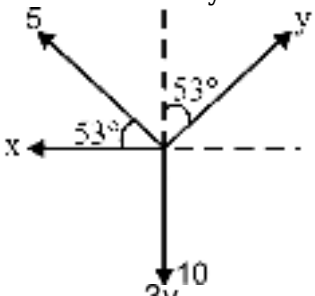
16) $\Sigma F_x = 0$

$\Rightarrow x + 5\cos53° - y\sin53° = 0$

$\Rightarrow x + 3 - \frac{4y}{5} = 0 \Rightarrow 5x + 15 - 4y = 0$

$\Sigma F_y = 0$

$10 - 5\sin53° - y\cos53° = 0$



$\Rightarrow 10 - 4 - \frac{3y}{5} = 0 \Rightarrow 6 = \frac{3y}{5} \Rightarrow y = 10$

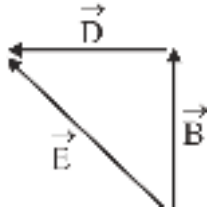
$\therefore$ x = 5.

17)

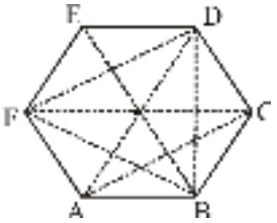
$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

18) Resultant may be zero if sum of three smaller displacement ≥ fourth displacement

19)



$\vec{B} + \vec{D} = \vec{E}$

20) 

$\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} + \overrightarrow{DE} + \overrightarrow{EF} + \overrightarrow{FA}$

$= \left(\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD}\right) + \left(\overrightarrow{DE} + \overrightarrow{EF} + \overrightarrow{FA}\right)$

$= \overrightarrow{AD} + \overrightarrow{DA}$

$= \overrightarrow{AD} - \overrightarrow{AD}$

$= \vec{0}$

21) $(X + Y)^2 + (X - Y)^2 + 2(X + Y)(X - Y)\cos\theta$

$= X^2 + Y^2$

$(X^2 + Y^2) + 2(X^2 - Y^2)\cos\theta = 0$

$\Rightarrow \cos\theta = -\dfrac{(X^2 + Y^2)}{2(X^2 - Y^2)}.$

22)

$m = \dfrac{f}{f - u} \Rightarrow -2 = \dfrac{-20}{-20 - u} \Rightarrow u = -30$ cm

23) Focal length of spherical mirror is half of radius of curvature, it does not depend on medium. There is phase shift of π in case of reflection of light from denser medium.

24) In the first case. Let x be the distances of object from the mirror. Then

$u = -x$

$v = +2x$

put f to –f

using $\dfrac{1}{v} + \dfrac{1}{u} = \dfrac{1}{f}$

or $\dfrac{1}{2x} - \dfrac{1}{x} = -\dfrac{1}{f}$ or $x = \dfrac{f}{2}$

In the second case, let y be the distance of object from the mirror. Then

$u = -y$, $v = -2y$

and put f to –f

So $\dfrac{1}{-2y} - \dfrac{1}{y} = -\dfrac{1}{f}$

Therefore $y = \dfrac{3}{2}f$

So, object will have to be moved by a distance of y-x or f.

25)

$m_T = -\frac{(-20)}{10} = -2$

$dv = m_T^2 du$

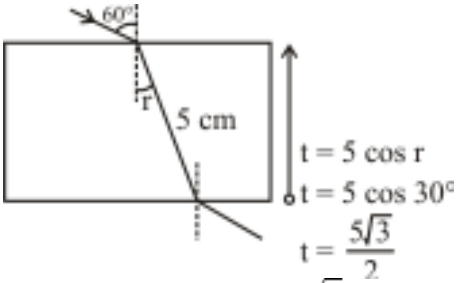
$= -(2)^2\ 0.1 = -0.4$ cm

0.4 cm away from the mirror.

26)

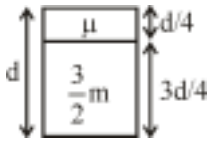
$\mu_{glass} = \sqrt{3}$ i = 60°

1 sin 60° = $\sqrt{3}$ sin r ; r = 30°



$d = \frac{t\sin(i-r)}{\cos r} \Rightarrow d = \frac{5\sqrt{3}}{2}\frac{\sin 30°}{\cos 30°}$

$d = \frac{5\sqrt{3}}{2} \times \frac{1}{\sqrt{3}} \Rightarrow d = \frac{5}{2}$cm



27)

$d_{a_{PT}} = d_{a_{P_1}} + d_{a_{P_2}}$

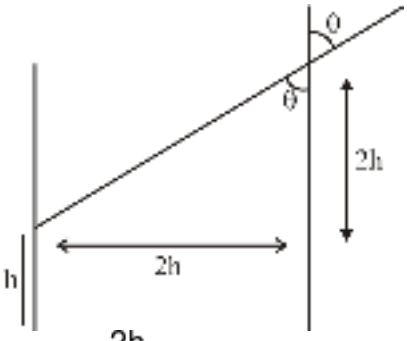
$\frac{d}{2} = \frac{d}{4\mu} + \frac{3}{4} \times \frac{2d}{3\mu} = \frac{d}{4\mu}[1+2]$

$\Rightarrow \frac{1}{2} = \frac{3}{\mu} \Rightarrow \mu = \frac{3}{2}$

28) **Explain the Question :** To find the refractive of the liquid if the beaker is filled with liquid up to a height 2h.

Concept : This question is based on apparent depth and heigh in refraction from plane surface.

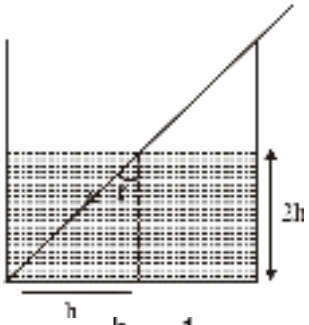
Solution :



$\tan\theta = \frac{2h}{2h} = 1$

θ = 45°

When beaker filled to 2h



$$\tan r = \frac{h}{2h} = \frac{1}{2}$$

$$\tan r = \frac{1}{2}$$

$$\sin r = \left(\frac{1}{\sqrt{5}}\right)$$

Using lenz law : $1 \times \sin 45° = \mu \times \sin r$

$$\Rightarrow \quad \mu = \left(\sqrt{\frac{5}{2}}\right)$$

Final Answer : 2

29) **Explain the Question :** To find image position after refraction for on object 'O' placed in front of glass.

Concept : This question is based on refraction spherical surface.

Solution :

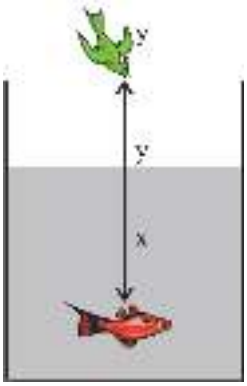
$$\frac{1.5}{V} - \frac{1}{-15} = \frac{0.5}{30}$$

$$\frac{1.5}{V} = \frac{1}{60} - \frac{1}{15} = \frac{-3}{60}$$

V = – 30 cm

Final Answer : 1

30) Here $x_{bf} = \mu y + x$



$$\frac{dx_{bf}}{dt} = \mu\frac{dy}{dt} + \frac{dx}{dt} = \frac{4}{3}(-12) + 24$$

$\Rightarrow \quad v_{bf} = 8$ m/s upward

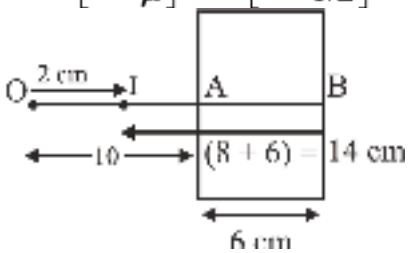
31)

Question Explanation:
The question asks for the apparent position of an object's image as seen through a glass slab, given the object's distance from the slab, the slab's thickness, and its refractive index.

Concept:
Apparent Shift in Glass Slab

Solution:
Shift of image towards slab

$$d = t\left[1 - \frac{1}{\mu}\right] = 6\left[1 - \frac{1}{3/2}\right] = 6\left[\frac{1}{3}\right] = 2\ \text{cm}$$



I will be formed at 14 cm for observer at B.

Final Answer: Option(3)

32)

Explain the Question : An object is placed at 21 cm in front of a concave mirror of radius 10 cm. A glass slab of thickness 3 cm and refractive index μ = 1.5 is placed at 15 cm from the mirror. Considering one refraction through the slab and one reflection from the mirror, find the final image position.
Concept : This question is based on refraction through slab and mirror reflection.
Formula : Shift due to slab = $t(1 - \frac{1}{\mu})$ Mirror formula: $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$, where $f = -\frac{R}{2}$
Solution : Shift due to slab = $3(1 - \frac{1}{1.5}) = 3(1 - \frac{2}{3}) = 3(\frac{1}{3}) = 1$ cm
Effective object distance from mirror = 21 – 1 = 20 cm
Mirror focal length $f = -\frac{10}{2} = -5$ cm
Using mirror formula : $\frac{1}{v} + \left(\frac{1}{-20}\right) = \left(\frac{1}{-5}\right)$

$$\frac{1}{v} = \left(\frac{1}{-5}\right) + \left(\frac{1}{20}\right)$$

$$= \left(\frac{-4+1}{20}\right) = \left(\frac{-3}{20}\right)$$

$v = \left(\frac{-20}{3}\right)$ cm (image on same side as object)
Final answer : Option (3)

33) **Asking About :** To find the optical angle of two media m_1 and m_2 for which speeds of light in m_1 and m_2 are given:
Concept : Total internal reflection

Formula : $\sin i_C = \frac{\mu_R}{\mu_D} = \frac{C_D}{C_R}$

Solution : Given speed of light in m_1

$C_R = 2.0 \times 10^8$ m/s, $C_D = 1.5 \times 10^8$ m/s

Calculation : $\sin i_C = \frac{1.5 \times 10^8}{2 \times 10^8} = \frac{3}{4}$

$\Rightarrow \left| i_c = \sin^{-1}\left(\frac{3}{4}\right) \right.$

Angle of incidence $i > i_C$

Final Answer option (3)

34) **Explanation:**

To find out the value of Angle 'θ' or sinθ, if ray of light totally reflected from AC to BC.

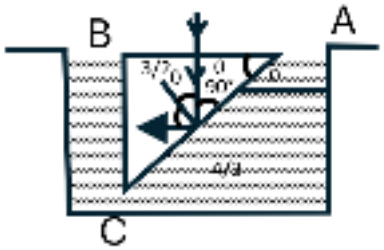
Concept:

TOTAL INTERNAL REFLECTION

formula

$\sin \geqslant \frac{\mu r}{\mu D}$

visual aid



As in given figure for y ray of light gets totally reflected from point the T.I.R at p takes place.

so, $\sin\theta \geqslant \frac{\mu R}{\mu D}$

or $\sin\theta \geqslant \frac{413}{312} \Rightarrow \sin\theta \geqslant 819$

correct option: (1)

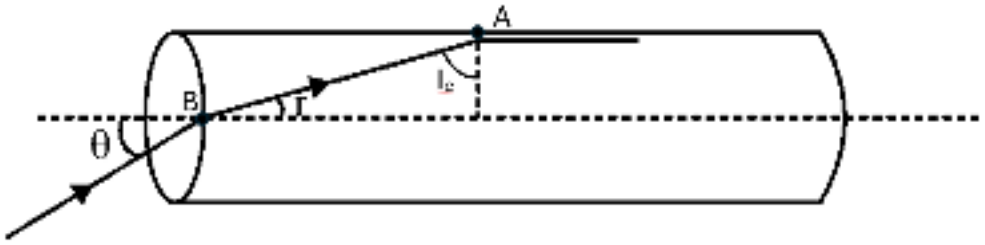
35)

Explain Question : A transparent cylindrical rod (refractive index = 2/√3) is in air. A light ray is incident on the mid-point of one end and, after refraction, grazes along the inner wall. Find the incident angle θ (in air).

Concept : This question is based on OPTICAL FIBRE

Formula : Snell's law: sin θ = n · sin r (n = refractive index of rod, r = angle inside rod)

Solution :



By snells law at point A → μ sin I_c = 1 sin 90º
By snells law at point B → 1 sinθ = μ sin (90º – I_c)
sinθ = μ cosI_c

$$\sin I_c = \frac{1}{\mu} \quad ...(1)$$

$$\cos I_c = \frac{\sin\theta}{\mu} \quad ....(2)$$

by square and add both the equation
$\mu^2 = 1 + \sin^2\theta$

$$\sin\theta = \frac{1}{\sqrt{3}} \quad (\mu = \frac{2}{\sqrt{3}})$$

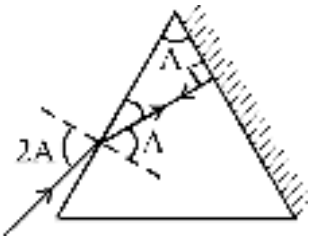
Final answer : Option (2)

36)

Here i = 0, r_1 = 0, r_2 = A = 30°
1.5 sin 30° = sin e

$$\Rightarrow \sin e = 1.5 \times \frac{1}{2} = 0.75$$

⇒ e = 48° 36'
δ = i + e – A = 0 + 48° 36' – 30° = 18° 36'

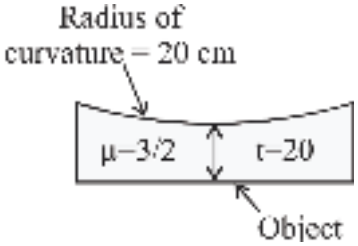
37) 

1 × sin2A = μ × sinA ⇒ μ = 2cosA

38) **Question Question :** Blue, green and red rays are incident normally on the vertical face of a right-angled prism. Inside the prism all rays strike the hypotenuse at an angle of incidence 45°.
Critical angles are :
Blue = 43°, Green = 44°, Red = 46°
We must determine which colours get separated.
Concept : This question is based on total internal reflection (TIR) and critical angle condition.
Formula :
For total internal reflection :
i > c
where

i = angle of incidence,
c = critical angle
Solution :
Inside the prism, angle of incidence on the hypotenuse :
i = 45°
Now compare with critical angles :
Blue :
$c_1 = 43°$
45° > 43° ⇒ TIR occurs
Green :
$c_2 = 44°$
45° > 44 ⇒ TIR occurs
Red :
$c_3 = 46°$
45° < 46° ⇒ No TIR, ray comes out of prizm
Thus :
Blue and green undergo total reflection
Red emerges out of the prism
So red gets separated from blue and green.
Final Answer: Option (3)

39) Radius of curvature = 20 cm



Considering refraction at the curved surface,
u = –20; $\mu_2 = 1$
$u_1 = 3/2$; R = +20
Applying $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$

$$\frac{1}{v} - \frac{3/2}{-20} = \frac{1-3/2}{20} = v = -10$$

i.e. 10 cm below the curved surface or 10 cm above the actual position of flower.

40)

$$m = \frac{v}{u} = -5 \text{ (real, inverted)}$$

⇒ v = –5 v
from $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

$$\Rightarrow \frac{1}{30} = -\frac{1}{5u} - \frac{1}{u} = \frac{-6}{5u}$$

⇒ u = –36 cm

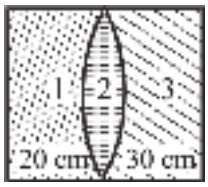
41) D = 80, x = 20

$$f = \frac{D^2 - x^2}{4D} = \frac{80^2 - 20^2}{320} = 18.75 \text{ cm}$$

42) $$\frac{1}{f_a} = (\mu_2 - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$
$$\frac{1}{f_m} = (_m\mu_g - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

43)

$P_{net} = P_1 + P_2 = 10D$
$P_1 + P_2 - dP_1P_2 = 6$
$10 - \frac{1}{4}(P_1P_2) = 6$
$P_1P_2 = 16$
$P_1 = 2D$; $f_1 = 0.5$ m
$P_2 = 8D$; $f_2 = 0.125$ m

44) 

focal length of the system
$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3}$$
Here,
for lens 1 ; $$\frac{1}{f_1} = \left[\frac{3}{2} - 1\right]\left[\frac{1}{-\infty} - \frac{1}{20}\right] = -\frac{1}{40}$$
for lens 2 ; $$\frac{1}{f_2} = \left[\frac{4}{3} - 1\right]\left[\frac{1}{20} - \frac{1}{(-30)}\right] = +\frac{5}{180}$$
for lens 3 ; $$\frac{1}{f_3} = \left[\frac{3}{2} - 1\right]\left[-\frac{1}{30} - \frac{1}{\infty}\right] = -\frac{1}{60}$$
$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3} = -\frac{1}{40} + \frac{5}{180} - \frac{1}{60}$$
F = –72 cm

45) $$\frac{1}{f_{net}} = \frac{1}{20} - \frac{1}{60} = \frac{2}{60} = \frac{1}{30}$$
$f_{net} = 30$ cm
u = -30 cm
 v = ∞

CHEMISTRY

46) $-\frac{d[N_2O_5]}{dt}$ = Rate of decomposition.

	$2N_2O_5$	→	$4NO_2$	+	O_2
Initial	100 torr		0		0
After time 10 min	100 – 2x		4x		x

Total pressure = 100 – 2x + 4x + x

= 100 + 3x

$x = \frac{50}{3}$ torr

So, rate of decomposition = $\frac{2 \times \frac{50}{3}}{10} = \frac{10}{3}$ torr min^{-1}

47) $-\frac{1}{2}\frac{d[N_2O_5]}{dt} = +\frac{d[O_2]}{dt}$

$-\frac{d[N_2O_5]}{dt} = 2 \times \frac{0.032}{32}$ mol $hour^{-1}$

and $-\frac{d[N_2O_5]}{dt} = 0.002 \times 108 = 0.216$ g $hour^{-1}$

48)

	$2A$	+	B	$\xrightarrow{K}$	C	+	D
at t = 0	C_0		$2C_0$		0		0
at t = 30 min	C_0–2x		$2C_0$-x		x		x

at 30 min [C] = $\frac{C_0}{4}$ = x

[A] = [C^0–2x] = $C_0 - \frac{2C_0}{4} = \frac{2C_0}{4}$

[B] = [$2C^0$–x] = $2C_0 - \frac{C_0}{4} = \frac{7C_0}{4}$

$r = k[A][B]^2$

$= k \times \frac{2C_0}{4} \times \left(\frac{7C_0}{4}\right)^2 = 49k\frac{C_0^3}{32}$

49)

A: Question Explanation

The question requires evaluating an assertion about the effect of a catalyst on the reaction rate and a reason explaining the mechanism, then determining the correct option.

B: Concept

A. **Factors affecting rate of reaction**

C: Reasoning

A. **Assertion (A) is correct.** By definition, a positive catalyst is a substance that increases the rate of a chemical reaction without itself undergoing any permanent chemical change.
B. **Reason (R) is incorrect.** A catalyst does not increase the energy of the reactant molecules. Instead, it provides an alternative reaction pathway with a lower activation energy (E_a). By

lowering the energy barrier, a larger fraction of reactant molecules possess the necessary energy to cross the barrier, leading to an increased reaction rate.

Since the Assertion is correct but the Reason is incorrect, the correct option is 3.

D: Final Answer

Assertion is correct but Reason is not correct.

The correct option is **3**.

50)

Question Explanation: Match each reaction with its correct order of reaction.

Concept: Reaction order is determined experimentally and depends on the rate law, not the stoichiometric equation.

Solution:

(A) Thermal decomposition of HI on gold surface
→ Surface-catalyzed reaction; rate becomes independent of concentration.
→ **Zero order reaction (P)**

(B) Inversion of cane sugar in acidic medium
→ Water is present in large excess.
→ **Pseudo first order reaction (Q)**

(C) Hydrogenation of ethene
$C_2H_4 + H_2 \rightarrow C_2H_6$
Rate $\propto [C_2H_2]$
→ **First order reaction (R)**

(D) Decomposition of NH_3 on hot Pt surface at high pressure
→ Catalyst surface gets saturated.
→ **Zero order reaction (P)**

Hence,

A → P, B → Q, C → R, D → P

Final Answer: Option 3

51) **Question Explanation:** To find out order of reaction.

Concept: n^{th} Order reaction

Solution:
$r = k (P_A)^n$
$(P_A)_0$ 363 torr when 5% reactant reacted
$r_1 = 1.0$ torr sec^{-1} $p_1 = 363 \times 0.95$

When 33% reactant is reacted.
$r_2 = 0.5$ torr sec^{-1} $P_2 = 363 \times 0.67$

$$\frac{r_1}{r_2} = \left(\frac{P_2}{P_1}\right)^n \Rightarrow \frac{0.5}{1} = \left(\frac{363 \times 0.67}{363 \times 0.95}\right)^n$$

$$\Rightarrow \frac{0.5}{1} = \left(\frac{67}{95}\right)^n$$

$2 = (1.417)n$

$2 = (2)^{n/2}$

$$\frac{n}{2} = 1 \Rightarrow n = 2$$

Final Answer: Option (4)

52) **Question Explanation:**
We are asked to identify the correct statements(s) related to chemical kinetics concepts like molecularity,
activation energy, and collision theory.

Concept:
Elementary reactions, activation energy, and effective collisions.

Solution:
• Statement: For an elementary reaction, molecularity is equal to the order of the reaction. correct
• Statement 2: Activation energy is defined as the energy difference between the complex and the reactants. correct
• Statement 3: An effective collision requires sufficient energy (≥ activation energy) and proper
orientation. correct
Since all three statements are correct, the correct choice is "All of the above".

Final Answer: Option(4)

53) A) r = rate of rds
$= R [Cl_2] [H_2S]$
(B) $r = R [Cl_2] [HS^{-1}]$

$$= R[Cl_2]\, k_{eq} \frac{[H_2S]}{[H^+]}$$

$$= R[Cl_2]\, k_{eq} \frac{[H_2S]}{[H^+]}$$

54)

If the assertion is true but the reason is false

55)

Explain the question : We are to determine the rate law for reaction A + 5B + 6C → 3L + 3M using given initial rate data.
Concept : This question is based on determination of order of reaction using the initial rate method.
Solution : Let rate = $k[A]^m[B]^n[C]^p$
From Exp. (1) & (2): [A] is halved (0.02 → 0.01), rate halves ($2.08\times10^{-3} \rightarrow 1.04\times10^{-3}$).
⇒ m = 1
From Exp. (1) & (3): [B] doubles (0.02 → 0.04), rate doubles ($2.08\times10^{-3} \rightarrow 4.16\times10^{-3}$).
⇒ n = 1
From Exp. (1) & (4): [C] doubles (0.02 → 0.04), rate becomes four times ($2.08\times10^{-3} \rightarrow 8.32\times10^{-3}$).
⇒ p = 2
Hence, rate law = $k[A]^1[B]^1[C]^2$
Final Answer : 1

56) For zero order, $k = \frac{a}{2(t_{1/2})_A}$

For first order, $k = \frac{0.693}{(t_{1/2})_B}$

$$\frac{a}{2(t_{1/2})_A} = \frac{0.693}{(t_{1/2})_B}$$

$$\frac{0.693}{2(t_{1/2})_A} = \frac{0.693}{(t_{1/2})_B} \Rightarrow 2(t_{1/2})_A = (t_{1/2})_B$$

57) $t_{99.9\%} = 3 \times t_{90\%}$

58) NCERT (OLD) XIIth Part - I ; Pg.No.119, Que.4.3

59)

	A	→	2B
t = 0,	P_0		0
t = 10,	$P_0 - x$		2x
t = 20,	$P_0 - y$		2y
t = ∞,	0		$2P_0$

$2P_0 = 20$ or $P_0 = 10$
at t → 20 min
$P_0 - y + 2y = 16.4$
$y = 16.4 - P_0$
$= 16.4 - 10 = 6.4$

$$K = \frac{2.303}{t}\log\frac{P_0}{P_0 - y} = \frac{2.303}{20}\log\frac{10}{10 - 6.4}$$

$$K = \frac{2.303}{20}\log\frac{10}{3.6} = \frac{2.303}{20}\log\frac{100}{36}$$

$$K = \frac{2.303}{20} \times 2 \times \log\frac{10}{6} = \frac{2.303}{10}\log\frac{10}{6}$$

for t=0 to 10 min;

$$K = \frac{2.303}{10}\log\frac{10}{10-x}$$

$$\frac{2.303}{10}\log\frac{10}{6} = \frac{2.303}{10}\log\frac{10}{10-x}$$

$$\frac{10}{6} = \frac{10}{10-x} \Rightarrow x = 4$$

$P_{total} = P_0 - x + 2x = P_0 + x = 14$

60)

On increasing temperature fraction of activated molecules increases.

61) **Question Explanation:** For the reaction x → 2y, the graph shows the concentration of x decreasing and y increasing. At time t, both concentrations are equal
Concept : This question is based on Stoichiometry and rate equations.
Solution :
Mathematical calculation
Let initial [x] = a, change = –x
Then, [x] = a – x, [y] = 2x
At time t,
⇒ a – x = 2x ⇒ a = 3x ⇒ x = a⁄3
 [x] = a – x = a – a⁄3 = (2a⁄3)This is when concentration becomes 2⁄3 of initial value ⇒ t = $t_{1/3}$
Final Answer : option (4)

62)

Question Explanation : Find initial reaction rate from graph.

Concept : Second order reaction.

Solution :

For 2nd order reaction -
Slope of $\frac{1}{[A]}$ vs time graph = rate constant (K)
Given tan θ = 0.6 ⇒ K = 0.6
Intercept $OC = \frac{1}{[A]_0} = 2$, So $[A]^0 = 0.5$
Initial rate = $K[A]_0^2 = 0.6\ (0.5)^2 = 0.150$
Initial rate = 0.150 mol L^{-1} min^{-1}

Final Answer: option (2)

63) $\because\ t_{1/2} = \frac{1}{aK}$

$$\log t_{1/2} = \log\left(\frac{1}{a}\right) + \log\frac{1}{K}$$

$$\log t_{1/2} = -\log a + \log \frac{1}{K}$$

64)

$r = k(A)^x$
$x = 0$
$r = k$

65) The problem provides a relationship between the logarithm of the rate constant (log k) of a first-order reaction and the temperature (T). The equation given is:

$$\log k = 15.0 - \frac{T}{10^4}$$

We are asked to determine the pre-exponential factor (A) and activation energy (E) based on this equation, using the Arrhenius equation.

B. Given Data

The given equation is:

$$\log k = 15.0 - \frac{T}{10^4}$$

We need to find the values of the pre-exponential factor (A) and the activation energy (E).

C. Concept

The Arrhenius equation describes the temperature dependence of reaction rates:
where:

- **k** is the rate constant,
- **A** is the pre-exponential factor,
- **E_a** is the activation energy,
- **R** is the ideal gas constant (8.314 J/mol·K),
- **T** is the temperature in Kelvin.

Taking the logarithm (base 10) of both sides:

$$\log k = \log A - \frac{E_a}{2.303RT}$$

We now compare this equation with the given equation:

$$\log k = 15.0 - \frac{T}{10^4}$$

D. Mathematical Calculation

To solve for A and E_a:

1. Solving for A :

- From log A = 15.0, we get:

$A = 10^{15.0} = 10^{15}$

2. Solving for E_a:

- From $\frac{E_o}{2.303\,R} = 10^4$, we solve for E_a:

$E_a = 10^4 \times 2.303 \times R$

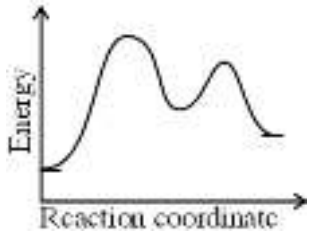
- Substituting R = 8.314 J/mol·K:

$E_a = 10^4 \times 2.303 \times 8.314 = 19147.342$ J/mol

- Converting to kJ/mol:

$E_a = \frac{19147.342}{1000}$ = 19.147 kJ/mol
• Rounding to one decimal place:
$E_a \approx 19.1$ kJ/mol

E. Final Answer
option 1.

66) 

67) **Explain Question:** Determine order of reaction from given half-life expression containing initial concentration.

Concept: General Integrated Rate Equation, Half-life dependence on initial concentration.

Solution:
For an nth order reaction, half-life depends on initial concentration as: $t_{1/2} \propto a^{-(n-1)}$
Given: $t_{1/2} = \frac{2\sqrt{k_1}a^{-2}}{4k_2 + 6\sqrt{5}}$ Thus $t_{1/2} \propto a^{-2}$ Comparing with general relation: $a^{-(n-1)} = a^{-2}$
n – 1 = 2
n = 3 Hence the reaction is third order.

Final Answer: Option (4)

68)

Explain Question:
We must match each each statement in Column I with the correct concept in Column II related to chemical kinetics.
Concept:
Arrhenius Equation
Solution:
Catalyst → lowers E_a
Molecularity → cannot be fractional or zero
First-order $t_1/_2$ → constant, independent of concentration
$e^{-E_a/RT}$ → fraction of molecules having $E \geqslant E_a$
Energetically favourable but slow → improper orientation
Area under MB curve → total probability = 1
A → R (catalyst lowers activation energy)
B →P (molecularity cannot be zero or fractional)
C → S (second half-life = first half-life for first-order reactions
D → U (fraction of molecules with $E \geqslant$ threshold energy)
E →Q (slow due to improper orientation)

F → T (total probability = 1)
So the matching order: A → R, B → P,C → S,D → U,E →Q,F →T
This corresponds to option 1.
Final answer: option (1)

69) $\log_{10}K_1 = \log_{10}A - \frac{Ea_1}{2.303\,RT}$ .... (1)

$\log_{10}K_2 = \log_{10}A - \frac{Ea_2}{2.303\,RT}$ .... (2)

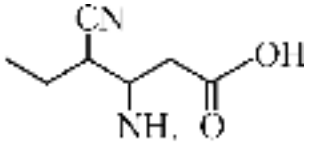
(2) – (1) $\Rightarrow \log_{10}\left(\frac{K_2}{K_1}\right) = \frac{Ea_1 - Ea_2}{2.303\,RT}$

$\Rightarrow \log\left(\frac{K_2}{K_1}\right) = \frac{24.9 \times 1000}{2.303 \times 8.314 \times 300} \Rightarrow \frac{K_2}{K_1} = 2.99 \times 10^4 ; 3 \times 10^4$

70)

NCERT Pg. No. # 103
Part-I
Topic - Integrated rate equation

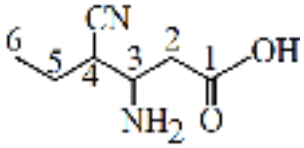
71) **B. Given Data:**



C. Concept:

A. **IUPAC Nomenclature for Organic Compounds with Multiple Functional Groups:** When a compound has more than one functional group, a priority order determines the parent name and the suffixes/prefixes used.
B. **Priority Order:** Carboxylic acids (-COOH) have higher priority than nitriles (-CN) and amines (-NH_2).
C. **Nitrile as a Substituent:** When a nitrile is not the principal functional group, it is named as a "cyano" substituent.
D. **Amine as a Substituent:** When an amine is not the principal functional group, it is named as an "amino" substituent.

D. Answer Explanation:



1. Identify the Main Functional Group: The compound contains a carboxylic acid (-COOH), a nitrile (-CN), and an amine (-NH_2). Carboxylic acids have the highest priority, so the parent name will be based on the carboxylic acid.
2. Number the Carbon Chain: Start numbering from the carboxylic acid carbon (which gets the number 1). The chain is six carbons long, making the parent chain "hexanoic acid."
3. Identify and Name Substituents:

A. At carbon 3, there is an amino group (-NH_2), making it "3-amino."

B. At carbon 4, there is a cyano group (-CN), making it "4-cyano."

4. Combine the Parts: Combine the substituent names alphabetically with the parent chain name: "3-amino-4-cyanohexanoic acid."
The correct IUPAC name is **3-amino-4-cyanohexanoic acid**
E. Final Answer: (Option 1).

72)

Explain Question : The problem asks which given pairs of compounds do **not** belong to the same homologous series.

Concept : Homologous series, Functional group classification, Organic compound families

Solution :

A. **Pair 1 (Alcohol vs Phenol):** → Different Functional Groups So, Not Homologous.

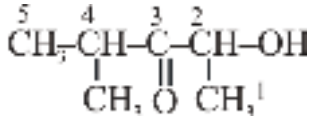
B. **Pair 2 (1°–Amine vs 2°–Amine):** → Different Functional Groups So, Not Homologous.

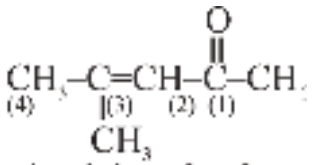
C. **Pair 3 (Acid vs Ester):** → Different Functional Groups So, Not Homologous.

Thus, all pairs do not represent the same homologous series.

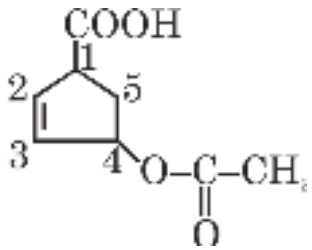
Final Answer : option (4)

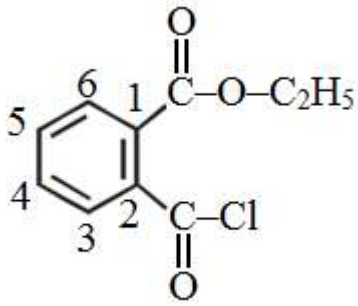
73) Functional groups present are aldehyde, amine, ketone, amide, carboxylic acid and acid anhydride.

74) 
2-Hydroxy-4-methyl pentan-3-one

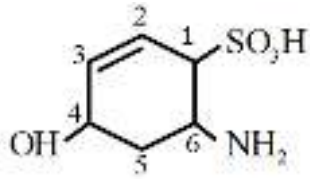
75) Mesityl oxide 

NCERT (XIIth) Part II, Pg. # 352, 366, 367

76) 

77) 

Ethyl–2(chlorocarbonyl)
Benzenecarboxylate

78) 

6-Amino-4-hydroxy cyclohex-2-ene-1-sulphonic acid

79)

IUPAC name is 3, 3-Diethyl-5-isopropyl-4-methyloctane.

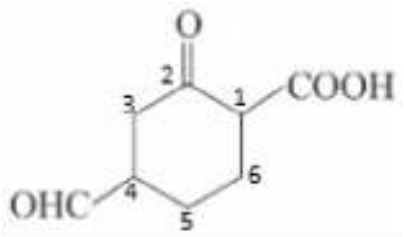
80) **Question Explanation:** Match each general formula with its correct compound type.
Concept: General formulas of homologous series.
Solution: Main Rule: Use standard formulas for organic families:

Formula	**Compound Type**	**Reason**
(i) C_nH_{2n-2}	Alkyne (c)	Matches Alkyne Formula
(ii) $C_nH_{2n}O_2$	Acid (a)	Matches carboxylic acid & Ester formula
(iii) $C_nH_{2n+2}O$	Alcohol (b)	Matches Alcohol & Ether formula
(iv) $C_nH_{2n+1}X$	Alkyl halide (d)	Matches Alkyl Halide Formula

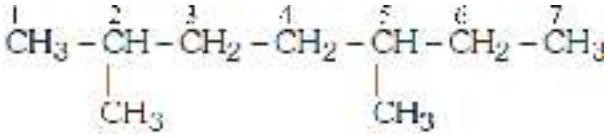
Correct match: a-ii, b-iii, c-i, d-iv
Final Answer: Option (2)

81) **Question Explanation:** Find the correct IUPAC name for a cyclohexane ring substituted with a carboxylic acid (–COOH), a ketone (= O), and an aldehyde (–CHO) group.
Concept: IUPAC nomenclature, Principal functional group, Priority order, Substituent numbering.
Solution:

A. **Identify Principal Group:** Carboxylic acid has the highest priority, so the parent name is cyclohexanecarboxylic acid.
B. **Numbering:** Start at the carbon attached to –COOH as C1. Number towards the ketone to give substituents the lowest possible locants.
    A. C1: –COOH (Carboxylic acid)
    B. C2: = O (Oxo group)
    C. C4: – CHO (Formyl group)
C. Alphabetical Order: "Formyl" comes before "Oxo".

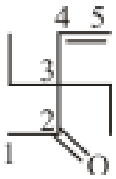


Final Answer: Option (3) (4-formyl-2-oxocyclohexanecarboxylic acid)



82)
2,5-Dimethyl Heptane

83) According to IUPAC



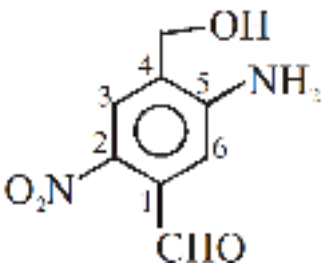
84)
3,3-Diethylpent-4-en-2-one

85)

Explain Question : IUPAC name of given compound is.

Concept : Is numbering is done in benzene ring (parent chain) such that prime pal functional group (–CHO) having ring carbon is given lowest number.

Solution :

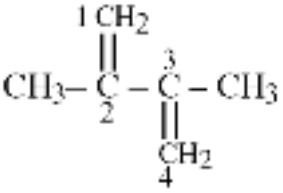
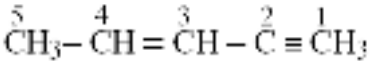
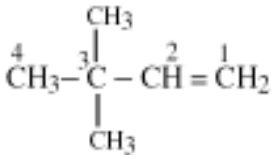
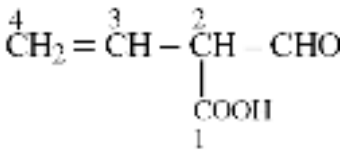


5-amino-4-hydroxymethyl-2-nitrobenzaldehyde
Numbering is done such that lowest set of locant rule is followed.
Compound is named as derivative of benzaldehyde.

Final Answer : option (4)

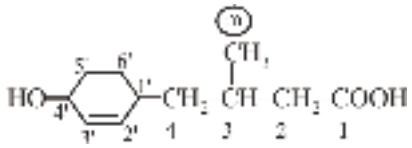
86)

Explain the Question : Identify which of the following IUPAC name is incorrect.
Concept : This question is based on IUPAC nomenclature.
Solution :

	Structure	**Given name**	**Incorrect /correct**
(1)		2,3-Dimethylbuta-1,3-diene	Correct name
(2)		Pent-3-en-1-yne	Correct name
(3)		3, 3-Dimethylbut-1-ene	Correct name
(4)		2-Vinyl-3-Oxopropanoic acid Correct name ⇒ 2-formylbut-3-enoic acid	Incorrect name

Final answer : 4

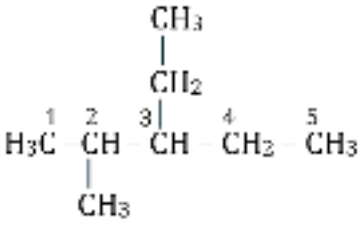
87) Acetone and acetaldehyde are F.G-isomers

88) 
4-(4-Hydroxycyclohex-2-enyl)-3-methylbutanoic acid

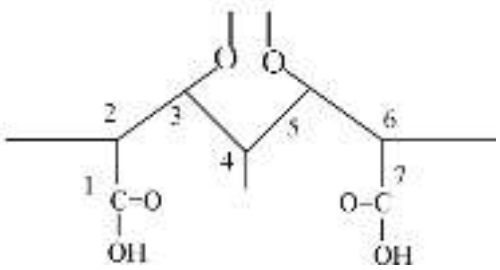
89) **Asking about :** Correct IUPAC name.

Concept : IUPAC naming rules

Solution/Explanation/Calculation :
• We have to first make structure of the compound from the given name (be it correct or incorrect). Then we have to name the compound whose structure is drawn, by following IUPAC Rules.



• 3-Ethyl-2-methyl pentane :
• Substituents are written alphabetically.
• lowest set of locant rule is followed while numbering.
Hence, the correct answer is option (3).

90) 

3,5-Dimethoxy-2,4,6-trimethyl heptan-1,7 dioic acid
OR
3,5-Di methoxy-2,4, 6-trimethyl heptane-dioicacid.

BIOLOGY

91) NCERT Based

92) NCERT-XI_Pg. # 109, 110, 111

93)

NCERT, Pg. # 112

94) NCERT, Pg. # 105

95) NCERT XI Page No. # 148

96) **Explaining the Question :-** Incorrect
Concept:- Ash analysis
Solution :-
The percentage weight of nitrogen in earth crust is very hight in comparison to human body (incorrect)Human body → Nitrogen more than earth crust
Final Answer :- (3)

97) **Question Explanation**
This question asks you to match different proteins with their correct descriptions or functions.

Concept
Biomolecules → Proteins → Types and functions of proteins
Explanation
Collagen is the most abundant protein in animals, GLUT-4 transports glucose into cells, Trypsin is a digestive enzyme, and RuBisCO is the most abundant enzyme in the biosphere.
Final Answer
Option 2 — A–III, B–IV, C–I, D–II.

98) NCERT, Pg. # 108, 109

99) NCERT-XI Pg # 118

100)

NCERT XI Eng. Pg. # 159, हिन्दी पेज # 159

101)

NCERT # XI, Page No. 158, Para - 9.12.5

102) NCERT XI Page # 116

103)

NCERT XI, Pg.# 117

104) NCERT Pg. # 154

105)

NCERT-XI, Pg. # 113

106) NCERT Pg. No.208

107) NCERT-XI, Pg. # 216 (3rd Para) & 217 (Fig.13.8), 219 (2nd para & Fig. 13.9) (E)
NCERT-XI, Pg. # 216 (आखिरी पेराग्राफ) & 217(चित्र 13.8), 219 (आखिरी पेराग्राफ एवं चित्र 13.9) (H)

108) NCERT Pg. # 209

109)

Explain Question: The question asks about the compound which is formed during photosynthesis.

Concept: Photorespiration.

Solution: The compound which is formed during photorespiration is: Phoshoglycolate.

So, the correct answer is: 2.

Final answer: Option 2

110) **Solution :** ATP synthesis in chloroplasts is called photophosphorylation, whereas ATP synthesis in mitochondria is called oxidative phosphorylation. Therefore, statement 4 is incorrect.
Final Answer : (4)

111) NCERT, Pg. # 137, 138

112) NCERT Pg. # 140,141

113) NCERT Pg.#139-145

114)

NCERT Page No. 209, (13.3)

115)

NCERT-XI, Page No.-217

116) **Explaining Question:**
This question asks about the direction of electron movement (uphill or downhill on redox potential scale) at different steps of the Z-scheme.

Concept:
Light reaction – Z-scheme and redox potential changes.
• Uphill = requires energy (light excitation).
• Downhill = spontaneous electron flow (ETS).

Solution:
Step A: PS-II → e^- acceptor
• Electron is excited by light → Uphill.

Step B: e^- acceptor → ETS → PS-I
• Electron flows through ETS → Downhill.

Step C: PS-I → e^- acceptor
• Again excited by light → Uphill.

Step D: e^- acceptor → $NADP^+$
• Electron moves downhill to reduce $NADP^+$ → Downhill.

Thus pattern = Uphill, Downhill, Uphill, Downhill.

Final Answer: Option 1

117) NCERT-XI, English Pg. # 216, 219
NCERT-XI, हिन्दी पेज # 216, 219

118) NCERT Based

119)

NCERT XI Pg. # 138

120) NCERT XI, Pg.No.#208

121) NCERT, Pg # 159

122) NCERT, Pg # 160

123) NCERT, Pg # 156

124) NCERT, Pg # 164

125) NCERT, Pg # 157

126) NCERT XI, Pg # 230 (Fig. 14.2)

127) NCERT Pg # 156

128) NCERT, Pg # 158,159,160,164

129)

Question Explanation
The question is asking which enzymes are responsible for converting pyruvic acid to ethanol and carbon dioxide.

Concept
Glycolysis.

Solution
Pyruvic acid is first decarboxylated to acetaldehyde and CO_2 by pyruvic acid decarboxylase, then acetaldehyde is reduced to ethanol by alcohol dehydrogenase. Both enzymes together

carry out the conversion from pyruvic acid to ethanol and CO_2.
Final answer: Option 1

130)

NCERT-XI, Pg. # 159

131) NCERT Pg. # 156,159,160

132)

NCERT Pg # 229

133) NCERT-XI, Pg. # 235

134) NCERT XI Pg.# 160

135)

NCERT XII, Pg.# 158

136) NCERT XI P. NO 180

137) NCERT, Pg # 169

138) NCERT, Pg # 176

139) NCERT, Pg # 176

140) NEW NCERT Pg. # 176

141) NCERT (XIth) Pg. # 245

142) NCERT (XI) Eng. Pg. # 173

143) NCERT Pg # 172, 173 (E)

144) NCERT Pg. # 175,176,177

145) NCERT, Pg # 176

146) NCERT Pg. No. 247, Para 15.4.1
ABA - growth Inhibitor
Ethylene-Included in both promotor and inhibitor but mainly in Inhinitor

147) NCERT, Pg # 177

148)

NCERT XI Pg.# 174, 175
[Inhibitor B, Abscission II and Dormin]

149) NCERT-XI, Pg. # 175,177,178

150) NCERT-XI, Pg. # 174, 175

151)

NCERT XI Pg. No. # 96

152)

NCERT XI Pg. No. # 91

153)

NCERT XI Pg. No. # 98

154)

NCERT XI Pg. No. # 95, 96

155)

NCERT XI, Pg # 122, 123, 124

156)

NCERT XI Pg. No. # 94

157)

NCERT XI Pg. No. # 96

158)

NCERT XI Pg. No. # 97, 98

159)

NCERT XI Pg. No. # 94

160) NCERT XI Pg. No. # 121

161) NCERT XI Pg. No. # 121, 122, 123, 124

162)

NCERT-XI, Pg. # 122-124

163)

NCERT XI Pg. No. # 120, 121

164) NCERT (XI) Pg # 100, 101

165)

NCERT XI Pg. No. # 125

166) NCERT-XI, Pg. # 69, 71

167) NCERT-XI, Pg.# 60

168) NCERT-XI, Pg. # 63, 64, 69

169) NCERT-XI, Pg. # 68

170) NCERT-XI, Pg. # 63, 66, 68
Tomato, potato, petunia, mustard, chinarose, mango and coconut

171) NCERT-XI, Pg. # 66, 69

172) NCERT-XI, Pg. # 64

173) NCERT-XI, Pg. # 68

174) (a), (b), (e)
NCERT-XI, Pg. # 64, 75, 76, 77

175) NCERT-XI, Pg.# 58

176) NCERT-XI, Pg. # 74

177) NCERT-XI, Pg. # 74

178) NCERT-XI, Pg. # 78, 79

179) NCERT-XI, Pg. # 75, 76

180) NCERT-XI, Pg. # 78